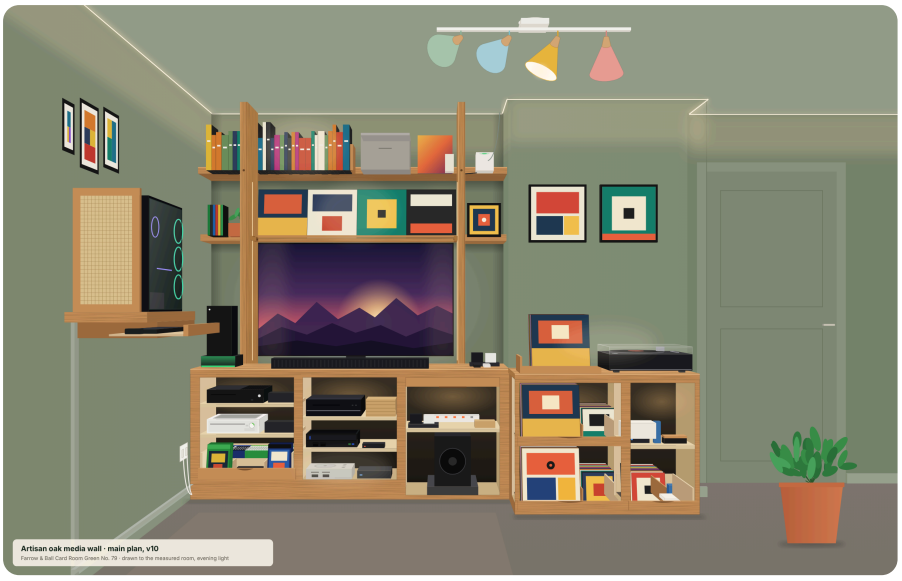
Home Network & Media Wall Plan

Revised Oct 18, 2026 · @emma · version 10, drawn to your measurements

The one big change. Your tape says the alcove is **186 cm wide and 38 cm deep** with a **235 cm** ceiling, not the 225 × 50 × 220 the last plan assumed. The TV, the Series X and the three bays below still fit; a turntable no longer does. So the whole vinyl corner (deck, LP browser and both face-out record trays) moves into a matching oak cabinet against the chimney breast, right beside the alcove. Everything else you chose stays.

1 · The plan on a page



Drawn to your measurements, in Card Room Green: the alcove, the vinyl cabinet against the chimney breast, the door to the hall beyond. Full-resolution renders come with this plan.

£2,207–4,249
plus the electrician, item by item in section 9

34 of 39
fit checks pass outright; 1 is tight, 2 need a check on site (section 3)

234 · 350
LPs and singles, all face-out

Area	Decision
Layout	Alcove: the TV between two oak uprights, the Series X on its fan stand to the left, a pad dock to the right; three 60 cm bays below for the Xbox consoles, the PlayStations and the sub. Chimney breast: an oak vinyl cabinet, 100 × 44 × 76 cm, deck and LP browser on top, record trays inside.
Records	Every LP face-out, about 234: 180 in two trays, 50 in the browser, 4 on the ledge over the TV. Every single face-out, about 350: 240 in the trays, about 110 in a new singles drawer.
Build	Artisan oak: solid oak uprights with wedged through-tenons and pegged housings, oak face frames, one 3 m solid oak worktop cut for both tops.
Heights	Worktop and cabinet top 76 · TV centre 116.5 · side shelves and ledge 156 · high shelf 196, 39 cm under the 235 ceiling.
Colour	Card Room Green No. 79, colour-drenched; skirting, door and trunking in Estate Eggshell.
Light	The ceiling LED strip as now, LED grooves in the uprights, a bias light, and a Wolston-style four-spot bar on the rose: sage and blue on the shelves, mustard on the coffee table, pink on the vinyl corner.
Network	Hub 5 in modem mode; Deco A and Switch 1 in bedroom 3; Cat6 down through the airing cupboard to Deco B and Deco C; spare ports for the projector and the music room.
Power	The side-wall socket becomes a double: the 10-gang behind the TV (A always on, B retro) and a switched 6-way in bay 3 that also feeds the deck and preamp.

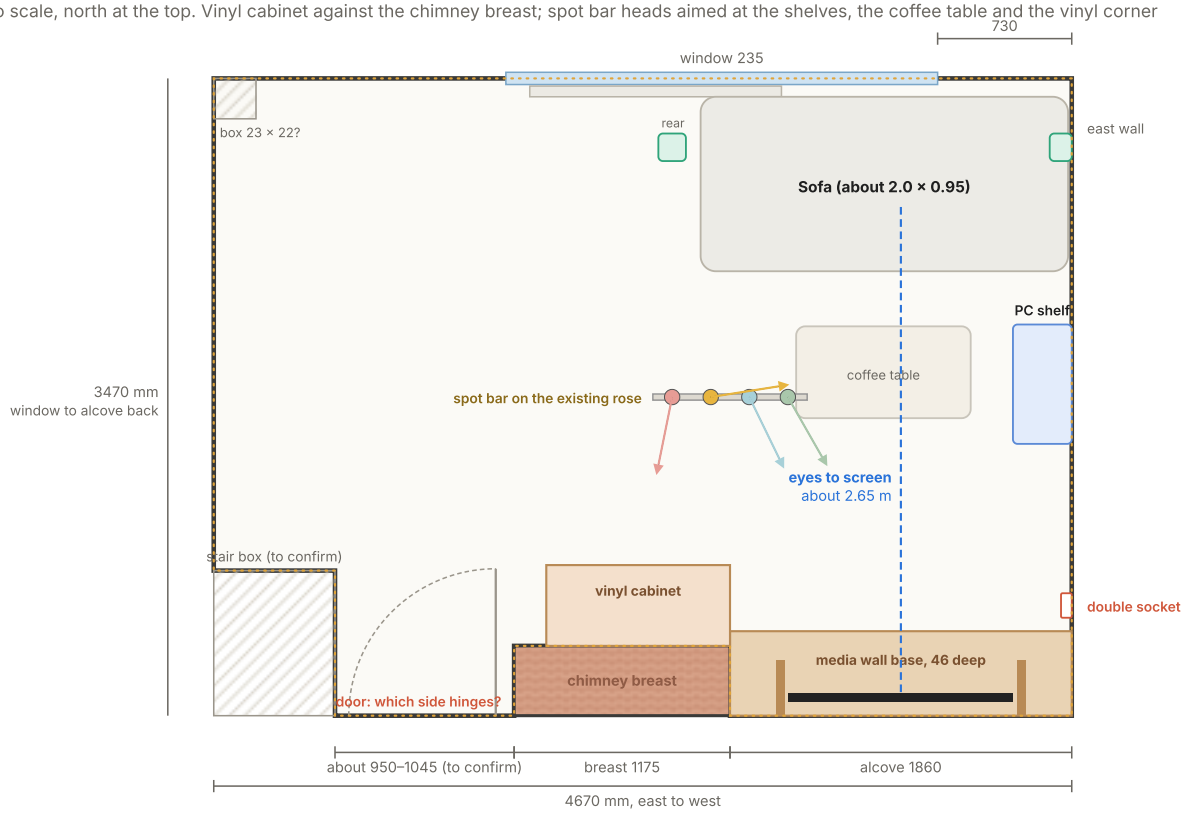
To confirm (full list in section 10): the vinyl corner moving to the chimney breast · where the door to the hall hinges, and the west end's sizes · the turntable, PC case and fan stand models, and the 10-gang's length · the 23 × 22 cm box by the window · wired ports for the projector and the music room.

2 · The room

Everything in this plan is now drawn from your hand-measured sketch. Where the agent's floor plan disagrees (it gives the living room as 4.95 × 3.65 m), your tape wins.

The living room from above, as you measured it

To scale, north at the top. Vinyl cabinet against the chimney breast; spot bar heads aimed at the shelves, the coffee table and the vinyl corner



Ceiling 2350 mm. Skirting 60 × 20. The west end (stair box, door, recess) is drawn roughly until you confirm it.

From your sketch	mm	Notes
Room, east to west along the window	4670	The media wall is the south wall, shared with the hall and garage
Window wall to the alcove's back wall	3470	To the chimney breast's face: 3090
Ceiling	2350	15 cm more than v9 assumed
TV alcove, east wall to the breast	1860 × 380	Floor to ceiling
Chimney breast	1175	Projects 380 on the alcove side, 390 on the west side
Window and radiator (north wall)	2350 · 1370	Window starts 730 from the east wall; radiator ends 130 inside its west end
Skirting	60 × 20	White now; Card Room Green eggshell in the plan
West end: recess, door, stair box	935 / 1045 / 660 / 820	Drawn roughly: please confirm which is which
Box in the window's west corner	230 × 220	Pipe boxing? To confirm

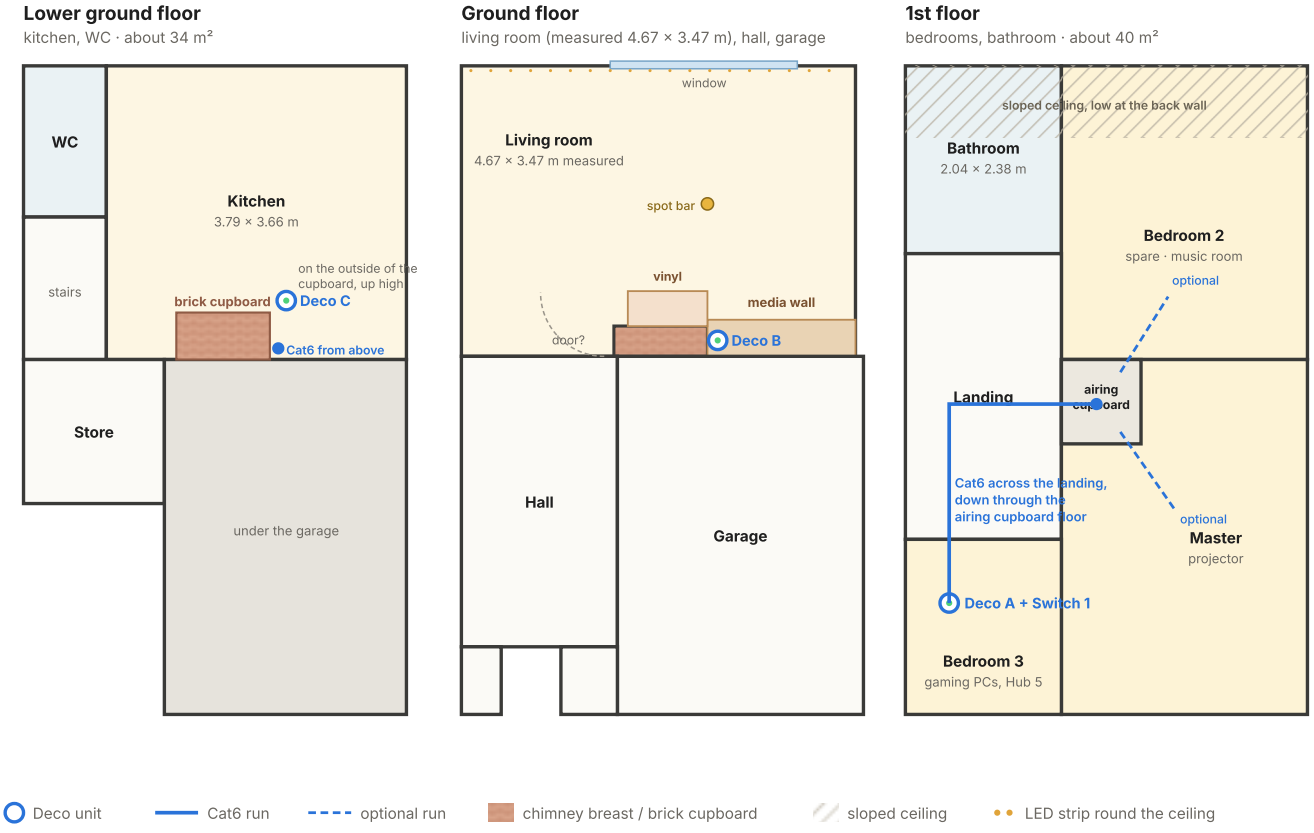
From the sofa: with the sofa against the window wall, 10 cm off the radiator, your eyes are about 2.65 m from the screen. A 55-inch fills about 26° of your view, which suits TV and games, and the screen's centre

at 116.5 cm sits about 1–2.5° above a seated eye line: comfortable. A 65-inch (about 145 cm wide) wouldn't fit between the uprights. The ceiling light hangs on the existing rose, in line with the breast's corner and 173.5 cm from the window wall; at 235 cm the lowest shade is at 212.6 cm, so headroom is no longer a worry.

The house

The house: where the Decos and Cat6 runs go

Deco A and Switch 1 move to bedroom 3 with the Hub 5. Both Cat6 runs drop through the airing cupboard, right over the media wall's corner



Outline traced from the agent's plan (north, the window side, at the top). The living room's alcove, chimney breast and windows are placed from your measurements.

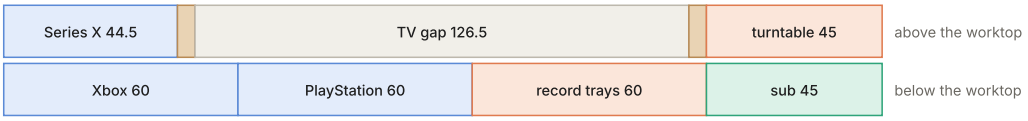
Deco A and Switch 1 now sit with the Hub 5 and the gaming PCs in bedroom 3. Both Cat6 runs go across the landing to the airing cupboard, which sits over the alcove's corner, and drop through its floor; section 7 has the details, including the optional runs to the master (projector) and bedroom 2 (music room).

3 · The media wall

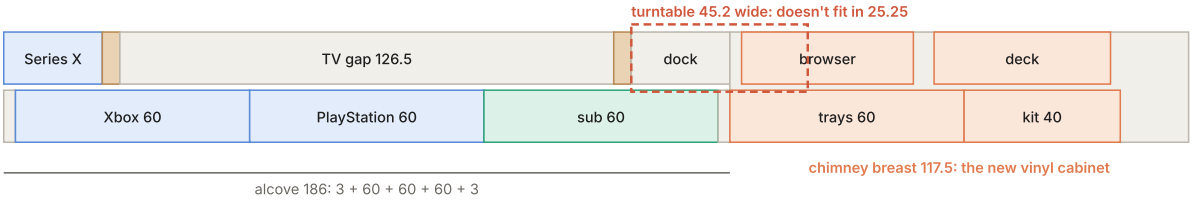
Why the layout changed

To scale, from the side wall. Top: what v9 assumed. Bottom: the room as you measured it

What we drew in v9: an alcove 225 wide



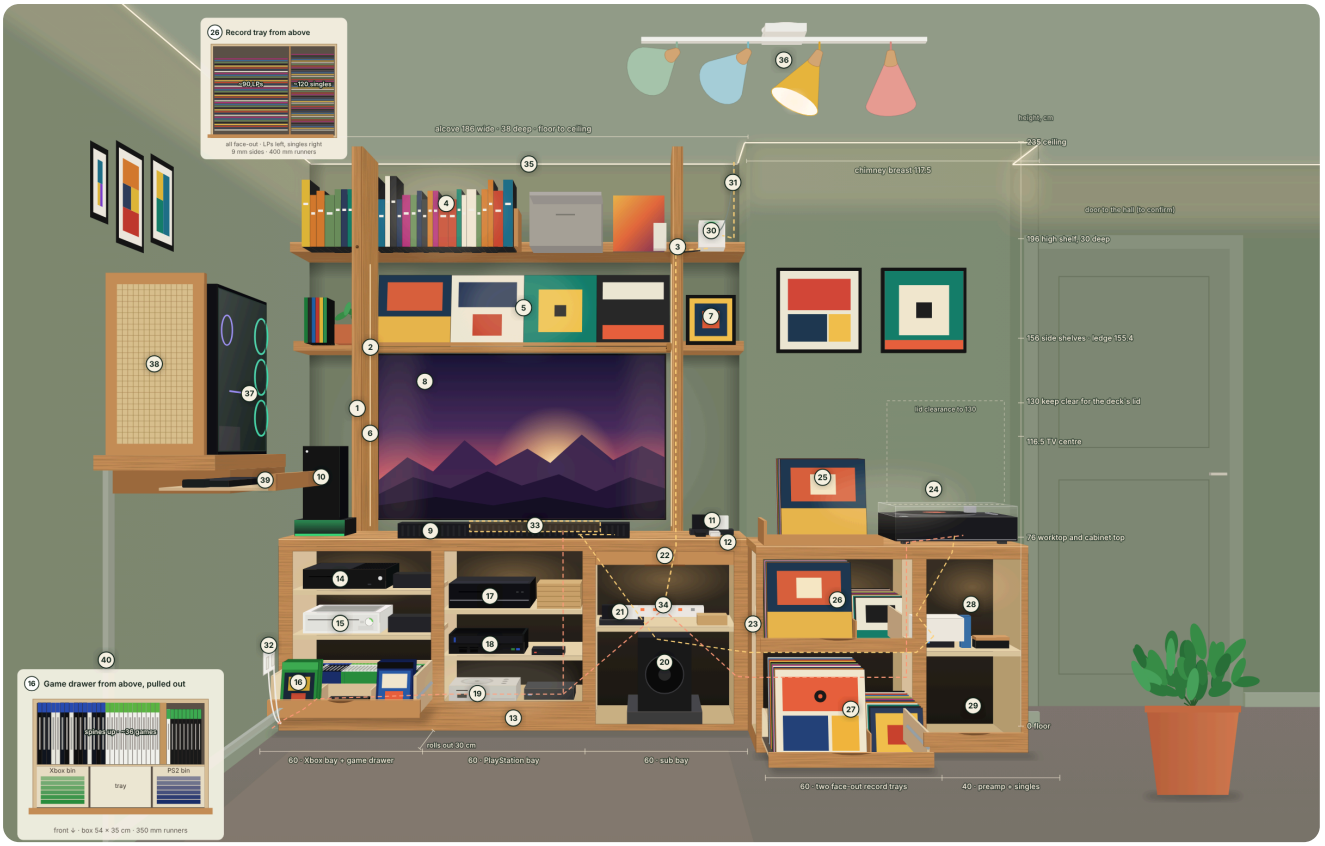
What you measured: 186 wide, then the chimney breast



- The alcove is 39 cm narrower than drawn. With the TV still framed by two uprights, each side is only 25.25 cm: enough for the Series X, not a turntable.
- Below, three 60 cm bays still fit. Bay 3 takes the sub (under the TV is good for bass), and the record trays move with the deck.
- The deck, the LP browser and both trays go into one oak cabinet against the chimney breast, at the same 76 cm height: the whole vinyl corner in one place.

Why this layout, and what we ruled out

- **Turntable on a shelf in the right-hand gap:** the gap is 25.25 cm; an AT-LP120X is 45.2 cm wide. Even a slim deck would overhang the upright and leave no room for the lid.
- **One upright, TV off-centre:** frees one wider gap, but loses the framed look, the second through-tenon and half the record ledge, and still leaves under 50 cm for a deck beside a 55-inch screen.
- **Series X inside a base bay:** it would lie flat, lose the fan stand and take the bay the sub needs.
- **Custom 36 cm carcasses flush with the breast:** a cleaner line, but the game drawer's 35 cm box would leave no room for plugs or a cable channel, and the BESTÅ frames couldn't be used. The base stands 8 cm proud of the breast instead, and the vinyl cabinet's oak end covers that step.



Numbered view: the game drawer and the lower record tray are pulled out 30 cm, with a view of each from above. Dashed lines show what's hidden: gold for data, audio and low-voltage leads, coral for mains. The numbers match the key below.

OAK JOINERY	
1	Solid oak uprights, 4.5 × 30, worktop to ceiling (159 long)
2	Wedged through-tenons: side shelves, top 156
3	Oak pegs where the high shelf and ledge are housed
4	High shelf, top 196: books up to 30 cm
5	Record ledge above the TV: 4 LPs face-out
6	LED groove in each upright washes the wall
7	Right shelf: a framed 7-inch, the “now playing” single
ON THE WORKTOP	
8	LG OLED C6 55in on its stand · centre 116.5
9	Soundbar · eARC, optical and AUX inputs
10	Xbox Series X on its fan stand
11	Pad charging dock and remotes
12	Solid oak worktop 186 × 46 × 3, top at 76
IN THE ALCOVE BASE	
13	Oak face frame; its bottom rail hides the 9 cm plinth
14	Xbox One + power brick · bay 1 top row
15	Xbox 360 + power brick · bay 1 middle row
16	Game drawer: 2 face-out bins, spine-up rows behind
17	PS3 + controller basket · bay 2 top row
18	PS2 + HDMI switch · bay 2 middle row
19	PS1 + RetroTINK · bay 2 bottom row
20	Subwoofer on a riser · bay 3
21	Shelf over the sub: pad dock, spares, 6-way behind
22	Hidden slim drawer under the worktop

THE VINYL CABINET ON THE CHIMNEY BREAST	
23	Oak cabinet 100 × 44 × 76, tight to the alcove
24	Turntable (AT-LP120X size allowed) · lid clear to 130
25	LP browser · ~50 LPs face-out
26	Upper record tray: ~90 LPs + ~120 singles
27	Lower record tray: the same, shown pulled out
28	Cubby: phono preamp, stylus, brush, sleeves
29	Singles drawer: ~110 singles face-out + cleaning kit
POWER, NETWORK AND LIGHT	
30	Deco B: leads in the groove behind the right upright
31	Cat6 from upstairs, through the ceiling
32	Double socket on the side wall, in front of bay 1
33	10-gang extension, hidden behind the TV
34	6-way switched strip at the back of bay 3
35	LED strip round the ceiling edge (as now)
36	Ceiling light: 4-spot cone bar on the existing rose
PC CORNER	
37	PC, glass side to the room · shelf top 110
38	Oak frame with woven cane hides the PC's ports
39	Keyboard + mouse tray, rolls out 25 cm
40	PC cables: shelf notch → trunking → skirting

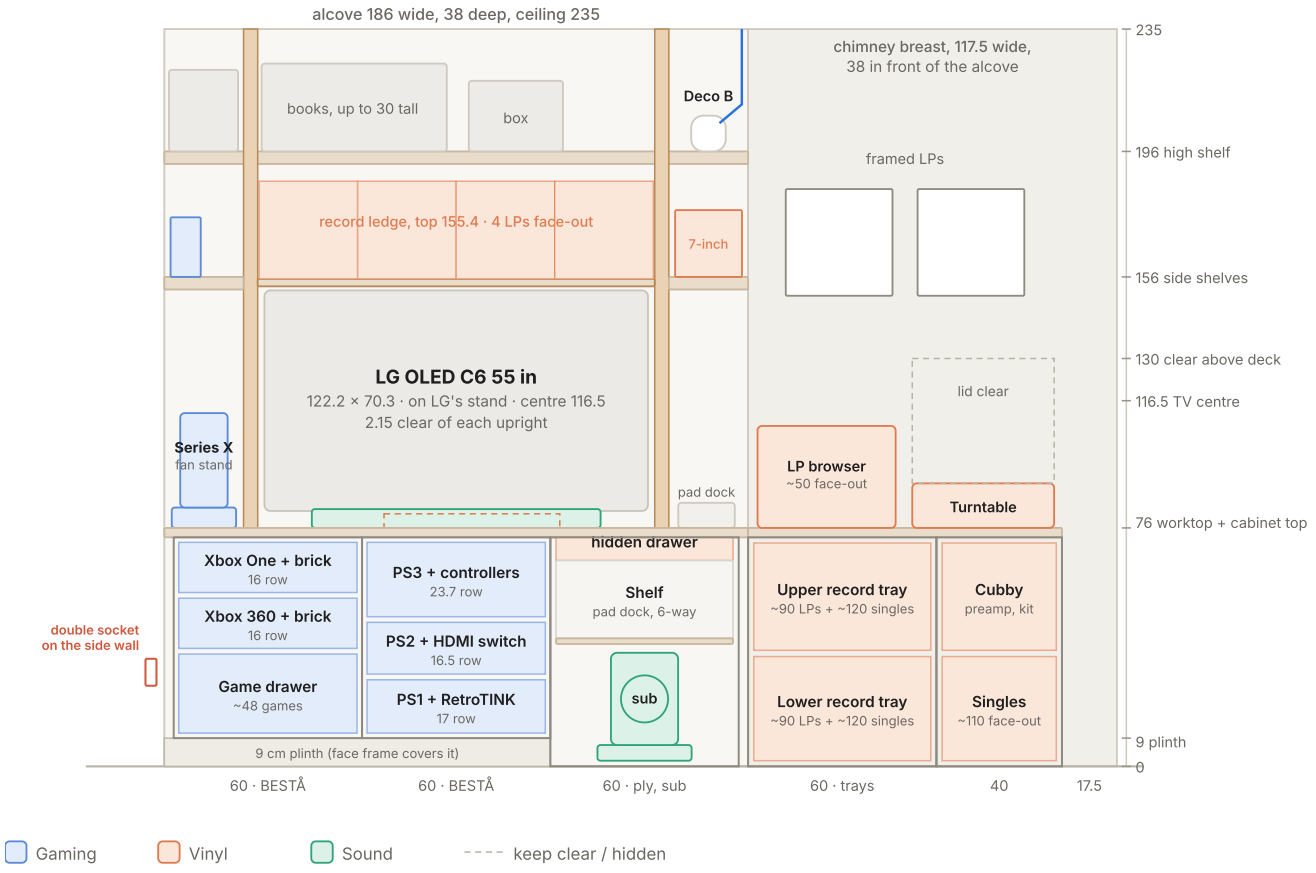
Built to your alcove: 186 wide, 38 deep, 235 to the ceiling

- **Base:** 3 cm filler, three 60 cm bays, 3 cm filler. Bays 1 and 2 are IKEA BESTÅ 60×40×64 frames on a 9 cm plinth (tops at 73, back panels off); bay 3 is an 18 mm ply carcass 60 × 40 × 73 standing on the floor. One oak face frame runs across all three; its bottom rail hides the plinth.
- **Depth:** 40 cm frames, a 4 cm cable channel behind them (flat two-compartment trunking) and the 2 cm face frame make 46 cm from the back wall, so the base stands 8 cm proud of the chimney breast. The vinyl cabinet's oak end panel closes that step on the right.
- **Worktop:** solid oak, 186 × 46 × 3, top at 76, with a brush-lined slot behind the TV for its leads and a grommet behind the Series X.
- **Oak uprights:** two, 4.5 × 30 cm, 159 cm from the worktop to the ceiling, at 25.25–29.75 and 156.25–160.75 cm from the side wall: 126.5 cm between them, 2.15 cm clear of the TV each side.
- **Either side of the TV:** 25.25 cm. The Series X stands on its fan stand on the left, with about 39 cm of air above it; the right side holds a pad charging dock and the remotes.

- **Shelves above:** oak side shelves at 156 either side, through-tenoned into the uprights; the record ledge at 155.4 between them, four LPs face-out; the high shelf at 196, wall to wall in three lengths, with books up to 30 cm, a box, a print and Deco B.
- **TV:** on LG's stand set 2 cm off the back wall: the screen's bottom at 81.4, centre 116.5, top 151.7, 1.3 cm under the ledge. The soundbar sits at the worktop's front edge; from the sofa you look over it.
- **Bay 3:** the sub on a 5 cm riser, facing the sofa, with 11.6 cm behind it for its rear port; a shelf above it for the pad dock and spares, with the switched 6-way at its back; a hidden slim drawer under the worktop.
- **Power:** the side-wall socket in front of bay 1 becomes a double; its two leads enter through a notch at the left end of the face frame's bottom rail and run back through the plinth to the channel.

The media wall from the front, to scale

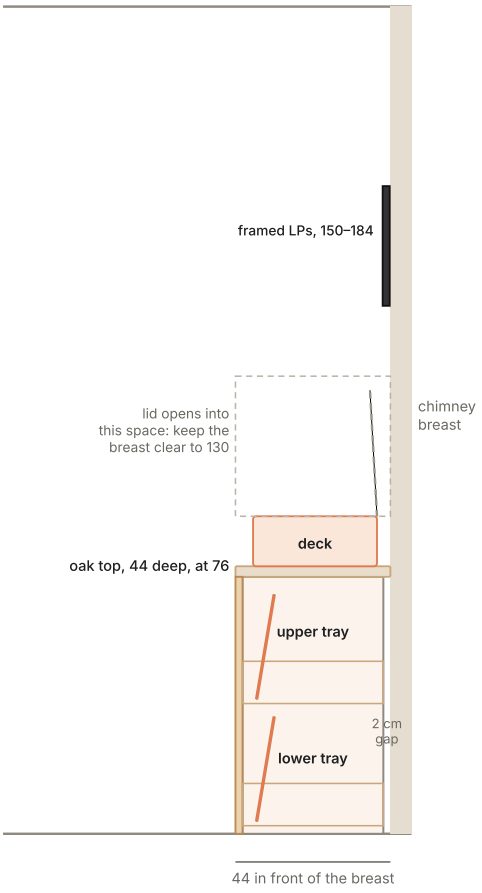
Oak uprights and pegged shelves above a 186 cm base in the alcove; the vinyl cabinet stands against the chimney breast. Heights in cm



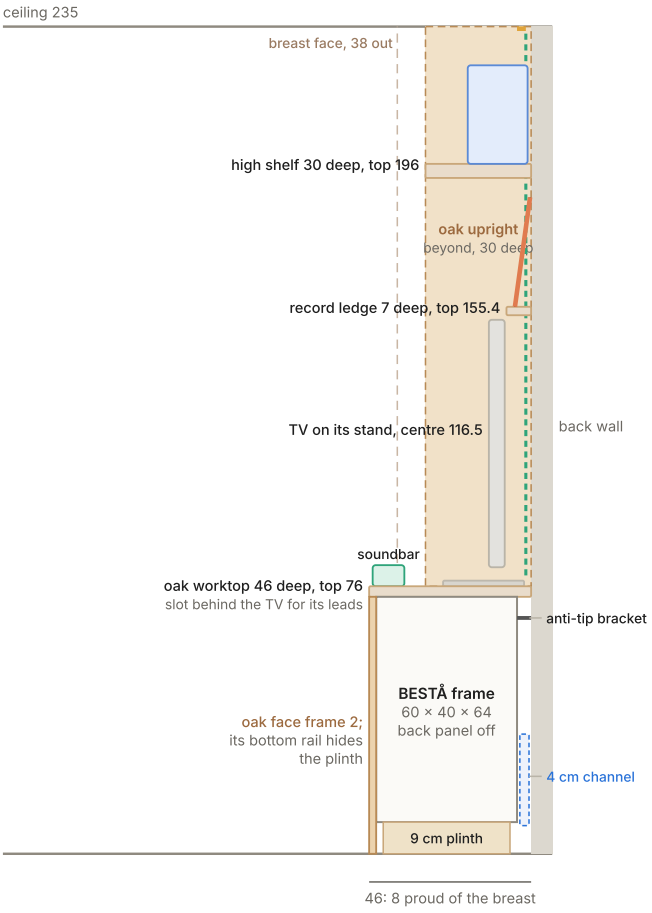
How the wall goes together, layer by layer

Side sections to scale, room on the left. Dashed: the upright and the breast's face beyond the cut

B · Through the vinyl cabinet



A · Through the middle of the alcove



The fit proof: every item against its published size

Each zone below lists what goes in it, its size and where that size comes from (numbered sources, section 10), the space, what's left and a verdict. 34 of the 39 checks fit outright. One is tight (the Series X's sides). Two need a check on site: the RetroTINK, whose maker publishes no size, and the door's swing past the vinyl cabinet. Two don't fit, and they're the reasons the layout changed.

Above the worktop, left of the TV

25.25 wide × 46 deep, worktop 76 to the side shelf at 152

Item	Size (cm)	Space	Spare (cm)	Result
Series X fan stand	20.5 × 20.5 footprint (assumed; model to confirm)	25.25 wide	2.38 each side	**Fits**
Xbox Series X, standing	15.1 × 15.1 × 30.1 tall [4]	25.25 wide	5.08 each side; Microsoft recommends 10–15 cm. The sides have no vents (air goes in underneath and out the top) and the zone is open at the front and above	**Fits, tight**
Clear above the Series X	top at 112.6 [4]	to 152	39.4 (Microsoft's minimum is 4)	**Fits**
Behind the Series X	stand 14 cm off the back wall [4]	46 deep	about 14.2 behind the console	**Fits**

The TV between the uprights

126.5 between the oak uprights; the record ledge's underside at 153

Item	Size (cm)	Space	Spare (cm)	Result
LG OLED C6 55 in, on its stand	122.2 × 70.3 × 4.5 panel; 75.7 tall on the stand [2]	126.5 wide	2.15 each side	**Fits**
Top of the screen	151.7 (bottom 81.4, centre 116.5) [2]	ledge at 153	1.3	**Fits**
LG stand base	47 × 23 [2]	worktop 46 deep	set 2 cm off the wall: 11 cm in front of it for the soundbar's leads	**Fits**
Soundbar, Hisense AX5100G	92 × 9 × 6 tall [3]	front edge of the worktop, top at 82	the line of sight to the screen's bottom edge passes 83.54–84.17 over it, above its top	**Fits**
A 65-inch TV instead	about 145 wide	126.5	about 18 short	**Doesn't fit**

Above the worktop, right of the TV

25.25 wide × 46 deep

Item	Size (cm)	Space	Spare (cm)	Result
Pad charging dock and remotes	about 18 × 9	25.25 wide	about 3.5 each side	**Fits**
The turntable (why it moved)	AT-LP120X: 45.2 × 35.2 [8]	25.25 wide	19.95 too wide	**Doesn't fit**

Shelves above

side shelves top 156, 25 deep; ledge top 155.4; high shelf top 196, 30 deep; ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Four LPs on the record ledge	4 × 31.4 = 125.6 wide[13]	126.5 long	0.9: snug, as drawn	**Fits**
Ledge LPs leaning back 8°	tops at 186.5[13]	high shelf underside 192	5.5 to lift one out over the 1.5 lip	**Fits**
Framed 7-inch on the right shelf	21.3 × 21.3 frame[13]	25.25 wide, 36 tall	2 each side	**Fits**
Books on the high shelf	up to 30 tall	39 to the ceiling	9 over the tallest; room for the ceiling strip's light	**Fits**
Deco B on the high shelf, right	11 × 11 × 11.4[7]	25.25 wide	7 each side, 27 clear above	**Fits**
Oak uprights	159 long (worktop to ceiling)[1]	235 ceiling	buy 165 and scribe to the ceiling	**Fits**

Bay 1, Xbox (BESTÅ frame on the 9 cm plinth)

56.8 wide inside, 40 deep, rows as in v9

Item	Size (cm)	Space	Spare (cm)	Result
Xbox One, original, and its brick	33.3 × 27.4 × 7.9; brick 17 × 7.5 × 4.8[12][11]	16 tall row	8.1 above; 21.5 beside the console for the brick	**Fits**
Xbox 360, original, and its brick	30.9 × 25.8 × 8.3; brick about 21 × 10 × 6[12][16]	16 tall row	7.7 above; 25.9 beside for the brick	**Fits**
Game drawer	box 54.3 × 35 on 350 mm runners; cases 19 × 13.5 × 1.4[15][13]	25.2 tall row, 40 deep	about 3 above the leaning cases; 5 behind the box	**Fits**
Depth with the leads	console 27.4 + HDMI plug about 5[12]	40 deep, back open	about 7.6, then the 4 cm channel	**Fits**

Bay 2, PlayStation (BESTÅ frame on the plinth)

56.8 wide inside, 40 deep

Item	Size (cm)	Space	Spare (cm)	Result
PS3, original, + controller basket	32.5 × 27.4 × 9.8[12]	23.7 tall row	13.9 above; 24.3 beside for the basket	**Fits**
PS2, original, + HDMI switch	30.2 × 18.3 × 7.8; switch 12.7 × 5.9 × 2.5[12][10]	16.5 tall row	8.7 above; 26.6 beside	**Fits**
PS1, original, + RetroTINK-6X CE	27 × 18.8 × 6; RetroTINK size not published[12][9]	17 tall row	29.8 beside the PS1 reserved for it; measure it when it arrives	**Check**

Bay 3, the sub (18 mm ply carcass from the floor)

56.4 wide × 40 deep inside, back open to the channel

Item	Size (cm)	Space	Spare (cm)	Result
Hisense subwoofer on a 5 cm riser	21.4 × 30.4 × 29.4 tall[3]	shelf underside at 39	top at 36.2: 2.8 above; 17.5 each side	**Fits**
Room behind the sub's rear port	port on the back[3]	46 from the front to the wall	11.6 to the wall	**Fits**
Shelf over the sub: pad dock, spares, 6-way	6-way about 34 × 6 × 4	56.4 × 36, top 40.8	24.8 clear up to the hidden drawer	**Fits**
Hidden slim drawer	front 56.4 × 7.4; box 52 × 30 × 6 on 300 mm runners[15]	65.6 to 73, under the worktop	—	**Fits**
Switch 2, TP-Link TL-SG108, on its edge in the channel	15.8 × 10.1 × 2.5[6]	4 cm channel	1.5	**Fits**

The vinyl cabinet on the chimney breast

100 × 44 × 76 against the 117.5 breast

Item	Size (cm)	Space	Spare (cm)	Result
The cabinet itself	100 wide, tight to the reveal[1]	breast face 117.5	17.5 to the breast's west corner for the door (to confirm)	**Check**
Two face-out record trays, bay A	trays 53.9 × 40 on 400 mm runners, 68–70 kg pairs[15][13]	carcass 56.4 inside, 69.4 tall	exact: 53.9 + 2 × 1.25 runner allowance = 56.4; finger room 2.6 and 2.1	**Fits**
Singles drawer, bay B	box 33.9 × 40; trough 18.9 for 18.3–18.4 sleeves[15][13]	36.4 inside	exact on the runners; 0.5 either side of the sleeves	**Fits**
Turntable + LP browser side by side on top	45.2 + 44 = 89.2 wide[8]	100 wide top	10.8 for gaps	**Fits**
Turntable depth, with room for its lid hinge	35.2 + 3.8 behind[8]	44 deep top	5 at the front	**Fits**
Lid open	deck top 90.2; lid about 35 deep[8]	breast kept clear to 130	frames start at 150	**Fits**

The room

467 × 347, ceiling 235

Item	Size (cm)	Space	Spare (cm)	Result
Eyes to screen from the sofa	sofa against the window wall, 10 off the radiator[1]	335 window wall to screen	about 2.65 m; a 55-inch fills about 26°	**Fits**
TV centre against seated eyes	116.5[2]	eyes at 105–112	about 1–2.5° above the eye line	**Fits**
Spotlight bar under the ceiling	22.4 drop[14]	235 ceiling	lowest shade at 212.6	**Fits**
PC shelf on the side wall	65 along the wall[1]	between the base and the sofa	about 29 clear of the sofa's front	**Fits**

Console sizes, every version

The bays are sized for the largest version of each console. The model number is on the label underneath; a smaller version just leaves more room.

Console	Version	W × D × H (cm)	Power
PS1	Original (largest)	27.0 × 18.8 × 6.0	Built in
PS1	PS one	19.3 × 14.4 × 3.8	External adapter
PS2	Original (largest)	30.2 × 18.3 × 7.8	Built in
PS2	Slim	23.1 × 15.2 × 2.8	External brick
PS3	Original (largest)	32.5 × 27.4 × 9.8	Built in
PS3	Slim · Super Slim	29.0 × 29.0 × 6.5 · 29.0 × 23.0 × 6.0	Built in
Xbox 360	Original (largest)	30.9 × 25.8 × 8.3	Large external brick
Xbox 360	S · E	about 27.0 × 26.4 × 7.5 · 25.4 × 26.4 × 6.7	External brick
Xbox One	Original (largest)	33.3 × 27.4 × 7.9	External brick
Xbox One	S · X	29.5 × 22.9 × 6.4 · 30.0 × 24.0 × 6.0	Built in
Xbox Series X	Standing	15.1 × 15.1 footprint, 30.1 tall	Built in

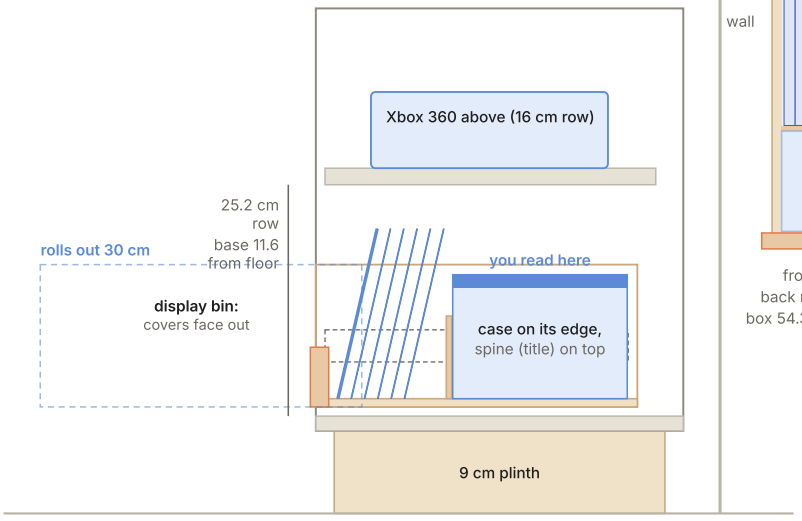
The game drawer (bay 1, bottom row)

Unchanged from v9. Two display bins at the front show covers face-out even with the drawer shut (Xbox left, PS2 right, a small tray between them); behind them the rest of the games stand on edge with the spine on top, so every title faces up when you roll it out.

The game drawer: covers on show at the front, every title readable from above at the back

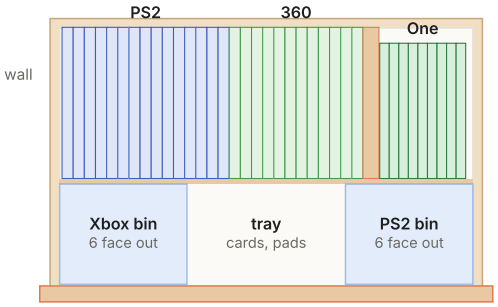
Bay 1, bottom row. Section left, plan right. Nothing stands behind anything else

Side section through bay 1



box 35 cm deep on 350 mm full-extension soft-close runners
5 cm free behind the box, then the 4 cm cable channel
bins: 19 cm cases leaning 13°, about 18.5 cm tall

Top view, drawer pulled out



front: oak lip with finger pull, matching the record drawer
back row: 15 PS2 + 12 Xbox 360 + follower block + 9 Xbox One
box 54.3 × 35 cm outside · 51.9 cm inside · 16 cm bins, 18.7 cm tray

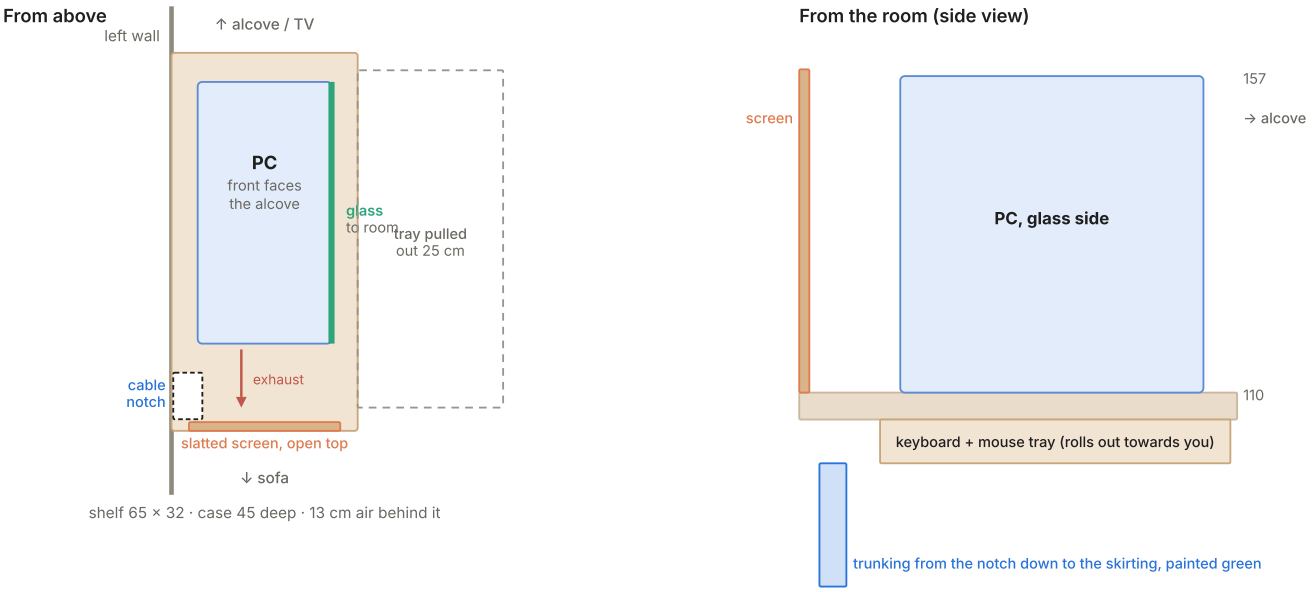
- **Box:** about 54.3 × 35 cm outside on 350 mm full-extension soft-close runners screwed to the BESTÅ sides; 51.9 cm inside; sides 15 cm; 9 mm base lined in felt.
- **Capacity:** about 48 games (12 face-out in the bins, about 36 spine-up), 5–6 kg of games and about 10 kg in all. Past about 48, keep the overflow spine-up in the box on the high shelf.
- **Cases:** PS2 and Xbox 360 19 × 13.5 × 1.4 cm; Xbox One 17 × 13.5 × 1.1–1.2 cm. Measure about ten of yours before cutting: cases vary by a few millimetres.
- **Runners:** Accuride DZ3832EC (43 kg) or Screwfix Essentials 964VN (40 kg); not IKEA's BESTÅ runners, which aren't full extension.
- **Cut list:** 12 mm birch ply sides 35 × 15.5 (2), back 51.9 × 15.5, front 51.9 × 6; 9 mm base 52.5 × 34.4 in a groove; 6 mm ply cross divider 51.9 × 9, bin walls 12.6 × 9 (2), raked bin floors 16 × 12.6 (2); oak follower blocks 19 × 12 (2); 20 mm oak lip 56.8 × 6.5.

The PC corner

The oak PC shelf stays on the side wall, moved 20 cm towards the alcove (148–213 cm from the alcove's back wall) so it clears the front of the sofa by about 29 cm.

The PC shelf: glass to the room, ports and cables screened from the sofa

A standard case has its window on the left, so with the glass facing the room the back ports face the sofa



- **Layout:** 65 cm along the wall, 32 cm out, top at 110. The case's glass faces the room; its back ports face the sofa behind an oak frame with a woven cane panel, open at the top for the exhaust.
- **Cables:** through a notch in the shelf's back corner, down slim trunking painted Card Room Green, and along the skirting into the base's channel. Over 3 m, use an active optical HDMI cable, "source" end at the PC.
- **Power:** ideally its own socket under the shelf (added on the electrician's visit), so the PC's 500–850 W stays off the media wall strips.
- **Keyboard and mouse:** a tray on 300 mm runners in a 7 cm housing under the shelf rolls 25 cm out towards the room.
- **Fixing:** a French cleat plus two steel brackets rated 25 kg or more; plugs and screws into masonry, or into studs on plasterboard.

4 · The vinyl corner

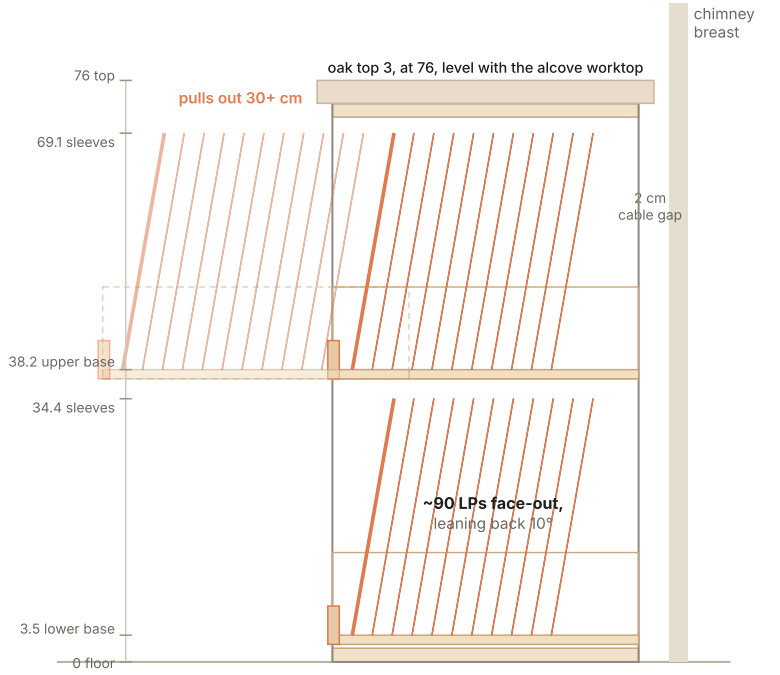
An oak cabinet against the chimney breast holds the whole vinyl collection and the deck in one place, at the same 76 cm as the alcove's worktop so the two read as one run of furniture. It's 100 cm wide, tight to the alcove, 44 cm deep and 76 cm tall: an 18 mm ply carcass behind an oak face frame, with an oak end panel on the side you see and a 2 cm gap at the back for its leads.

- **Bay A, 60 cm:** the two face-out record trays from v9, unchanged: about 90 LPs and 120 singles each, all face-out.
- **Bay B, 40 cm:** an open cubby for the phono preamp, spare stylus, brush, cleaning fluid and inner sleeves; below it, a new singles drawer with about 110 singles face-out beside the cleaning kit.
- **On top:** the LP browser (44 × 24, about 50 LPs face-out) on the left, the turntable on the right on isolation feet. The top takes a deck up to the size of an AT-LP120X (45.2 × 35.2 cm), with 3.8 cm behind it for the lid hinges.
- **Lid:** keep the chimney breast clear up to 130 cm so the lid opens fully. Above that, two framed LPs from 150 cm.
- **The door:** the cabinet stops 17.5 cm short of the breast's west corner. If the door to the hall hinges on the breast side and opens into the room, check its handle clears the cabinet's end before building.
- **Safety:** two loaded trays weigh about 32 kg each. Fix the cabinet to the breast with anti-tip brackets before loading it, and open one tray at a time.
- **Sound:** the deck is about a metre from the sub, on a separate cabinet, so bass doesn't feed back into the stylus; isolation feet help further.

The record trays: every LP and single face-out, two tiers in the vinyl cabinet

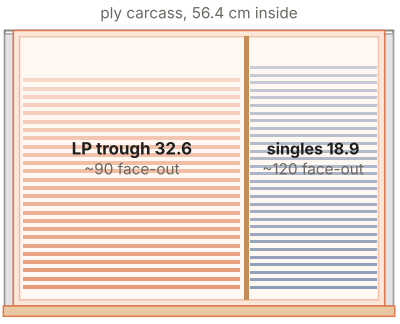
Bay A of the oak cabinet on the chimney breast. Section left, plan right. Pull a tray and flip through the covers like a record shop browser

Side section through bay A



carcass 18 mm birch ply, 60 × 40 × 73, back open · trays 40 deep on 400 mm runners
12 mm base, 9 mm sides, 5 cm oak lip · 2.6 and 2.1 cm finger room (top panel at 71.2)

Top view of one tray



tray 53.9 × 40 outside, 52.1 × 38.2 inside
9 mm sides · 6 mm oak divider · 5 cm oak lip
about 32 kg full: open one tray at a time

How the record trays work

- **LP trough:** 32.6 cm wide, so a bare 31.4 cm sleeve has 6 mm each side. Records lean back 10° like a record shop browser; the last one leans on a raked oak rest.

- **Singles trough:** 18.9 cm wide, the same lean, a smaller rest at the back.
- **Capacity:** 38.2 cm inside. About 31 cm of LP stack at about 3.3 mm each (gatefolds allowed for): about 90. About 34 cm of singles at about 2.8 mm: about 120. Two trays: about 180 LPs and 240 singles.
- **Outer sleeves:** if most LPs live in thick outer sleeves (up to 32.3 cm), widen the LP trough to 33.4 cm; the singles trough then drops to 18.1 cm, too narrow for face-out singles.
- **Runners:** 400 mm full extension, 68–70 kg per pair, screwed straight into the 18 mm ply.

Dimension	Figure	How it adds up
Bay A carcass	60 × 40 × 73 outside, 56.4 × 69.4 inside	18 mm birch ply; back open to the cable gap
Tray width	53.9 outside, 52.1 inside	56.4 opening − 2 × 1.25 runners; 9 mm sides
Tray depth	40 outside, 38.2 inside	9 mm front and back panels
Inside layout	32.6 + 0.6 + 18.9	LP trough, oak divider, singles trough
Lower tray	Underside 2.3, base top 3.5, sleeves to 34.4	Lip 2.3 to 7.3; 2.6 cm finger room under the upper tray
Upper tray	Underside 37.0, base top 38.2, sleeves to 69.1	Lip 37.0 to 42.0; 2.1 cm under the top panel at 71.2
Weight	About 32 kg per tray, full	90 × 0.23 kg + 120 × 0.045 kg + about 5 kg of tray

The singles drawer (bay B)

Bay B is 40 cm wide outside and 36.4 cm inside, which takes a 33.9 cm drawer box on the same 400 mm runners. One 18.9 cm trough holds about 110 singles face-out, leaning back 10°; the 13 cm beside it takes the cleaning kit and spare inner sleeves. Its 5 cm oak lip matches the trays. The cubby above it is open, so the preamp's light shows and its leads are easy to reach.

Cut list

Part	Material	Size (cm)	Qty
Carcass sides (ends and the shared middle)	18 mm birch ply	73 × 40	3
Top and bottom panels, bay A · bay B	18 mm birch ply	56.4 × 40 · 36.4 × 40	2 + 2
Back rails, bay A · bay B	18 mm birch ply	56.4 × 8 · 36.4 × 8	2 + 2
Bay B shelf	18 mm birch ply	36.4 × 38, top at 37	1
End panel (the side you see)	Oak-veneered ply or solid oak	76 × 44	1
Face frame	20 mm oak	100 × 73, stiles about 5 wide	1 set
Top	Solid oak, from the worktop board	100 × 44 × 3, chamfered front	1
Record trays (each)	12 mm ply base 53.9 × 40; 9 mm ply sides 40 × 12, back 52.1 × 12, front 52.1 × 4.5; 6 mm oak divider 38.2 × 10; raked rests; 15 mm oak lip 56.8 × 5; felt	—	2

Singles drawer	12 mm ply base 33.9 × 40; 9 mm ply sides 40 × 12, back 32.1 × 12, front 32.1 × 4.5; 6 mm oak divider; 15 mm oak lip 36.8 × 5	—	1
Runners	400 mm full extension, side mount, 12.5 mm; 68–70 kg pairs for the trays, 45 kg+ for the singles	—	3 pairs

How to build it

1. **Check the door** swings clear of where the cabinet's west end will be, and that the breast's face is flat (scribe the back edge if not).
2. **Build the carcass:** glue and screw the three sides, the top and bottom panels and the back rails, square; fit bay B's shelf at 37.
3. **Set it in place** tight to the alcove's base, shimmed level so its top meets the alcove worktop at 73, and fix it to the breast with anti-tip brackets through the 2 cm gap.
4. **Fit the runners:** the trays' lower pair so the tray underside sits at 2.3, the upper pair at 37.0; the singles drawer's at 2.3.
5. **Build and hang the trays and the drawer** (glue and screw the bases under the sides; glue the oak dividers down to stiffen them), then fit the oak lips 2–3 mm clear of the face frame.
6. **Fit the face frame, end panel and top,** oil the oak, and load one tray at a time.

Vinyl through the soundbar, done properly

- **Signal path:** turntable set to PHONO → external phono preamp in bay B's cubby → a 3 m RCA-to-3.5 mm lead along the cabinet's back gap, through bay 3 and the channel to the soundbar's AUX input.
- **Check first:** the external preamp only helps if the deck has a PHONO/LINE switch. With no switch, plug the deck straight into the soundbar's AUX.
- **Soundbar:** Music mode, bass up, surround, night and dialogue modes off. Always the AUX cable, never Bluetooth.
- **Stylus:** the exact replacement for the deck's model, tracking weight set to the cartridge maker's figure. Brush records before playing.
- **Upgrade path:** a stereo amp with a phono input and sub output plus passive speakers (about £200–450), or an AV receiver with phono and eARC in place of the soundbar (£500+).

Where every record lives

Where	What	How many
Vinyl cabinet, upper and lower trays	LPs and singles, face-out	About 180 LPs + 240 singles
Vinyl cabinet, singles drawer	Singles, face-out	About 110
On the cabinet	LP browser, face-out	About 50 LPs
Above the TV	Oak record ledge, the "now playing" wall	4 LPs
Right shelf	A framed 7-inch	1
Total		**About 234 LPs and 350 singles, every one face-out**

If the collection grows past that, a 2 × 4 cube unit (about 33 cm cubes, 39 cm deep) holds about 500 LPs on its side behind the sofa.

5 · The oak joinery

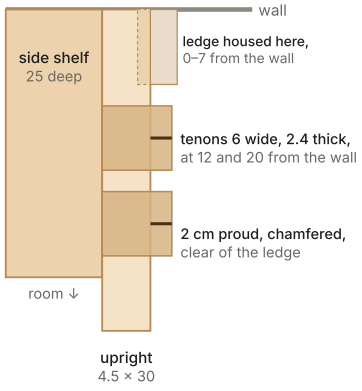
Above the worktop the wall is furniture rather than cladding: two solid oak uprights, shelves carried on joints you can see, and the Card Room Green wall left showing between them.

- **Two uprights frame the TV:** solid oak, 4.5 × 30 cm, 159 cm long from the worktop to the ceiling, each on two 10 mm dowels, scribed to the ceiling and held to the solid back wall with concealed brackets.
- **Wedged through-tenons:** each side shelf (top 156) passes through its upright on two tenons, 6 cm wide and 2.4 thick, showing 2 cm on the TV side with an oak wedge. They sit 12 and 20 cm from the wall, in front of the ledge's housing, so the two joints never meet.
- **Housed and pegged:** the high shelf, in three lengths at 196, and the record ledge at 155.4 are housed 1.2 cm into the uprights and pinned with 10 mm oak pegs, which show as dark dots.
- **Record ledge:** 7 cm deep with a 1.5 cm lip; four LPs lean against the wall up to about 186.5, 5.5 cm under the high shelf.
- **LED grooves:** a routed groove on each upright's inner face holds a 2700K strip that washes the wall either side of the TV.
- **Hidden cables:** a 2 × 1.5 cm groove down the back of the right upright carries Deco B's 12 V lead and patch lead to the worktop slot. A small notch at the top of each upright lets the ceiling LED strip run straight past.
- **Finish:** hardwax oil throughout; lightly limed or smoked oak on the uprights avoids the bright orange "new oak" look.

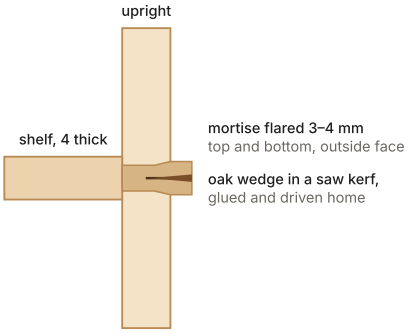
Artisan oak: how the joints work

Drawn to scale at 7 px per cm. The joints carry the load and are left on show, finished in hardwax oil

A · Wedged through-tenons, from above

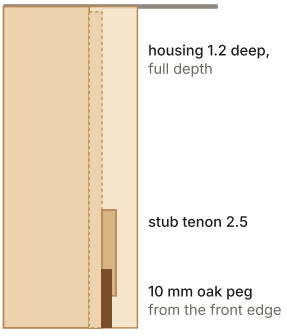


B · The same joint from the front



The kerf runs flat, so the wedge spreads the tenon up and down, along the upright's grain, and can't split it.

C · Housed shelf, pegged



high shelf / ledge
seen from above

• Through-tenons (A, B): both side shelves (top 156) pass through their upright, 2 cm proud on the TV side, 12 and 20 cm from the wall.

• Housed and pegged (C): the three lengths of the high shelf (top 196) and the record ledge (top 155.4). The peg shows as a dark dot on the upright's face.

• Uprights: 159 long, worktop to ceiling, each on two 10 mm dowels into the worktop, scribed to the ceiling, with two concealed brackets into the solid back wall.

Joinery cut list

Part	Material	Finished size (cm)	Joints	Qty
Uprights	Solid oak, quarter-sawn if you can	159 × 30 × 4.5 (buy 165 and scribe)	Two mortises each for the side shelf, housings for the high shelf and ledge, LED groove inside, cable groove at the back of the right one	2

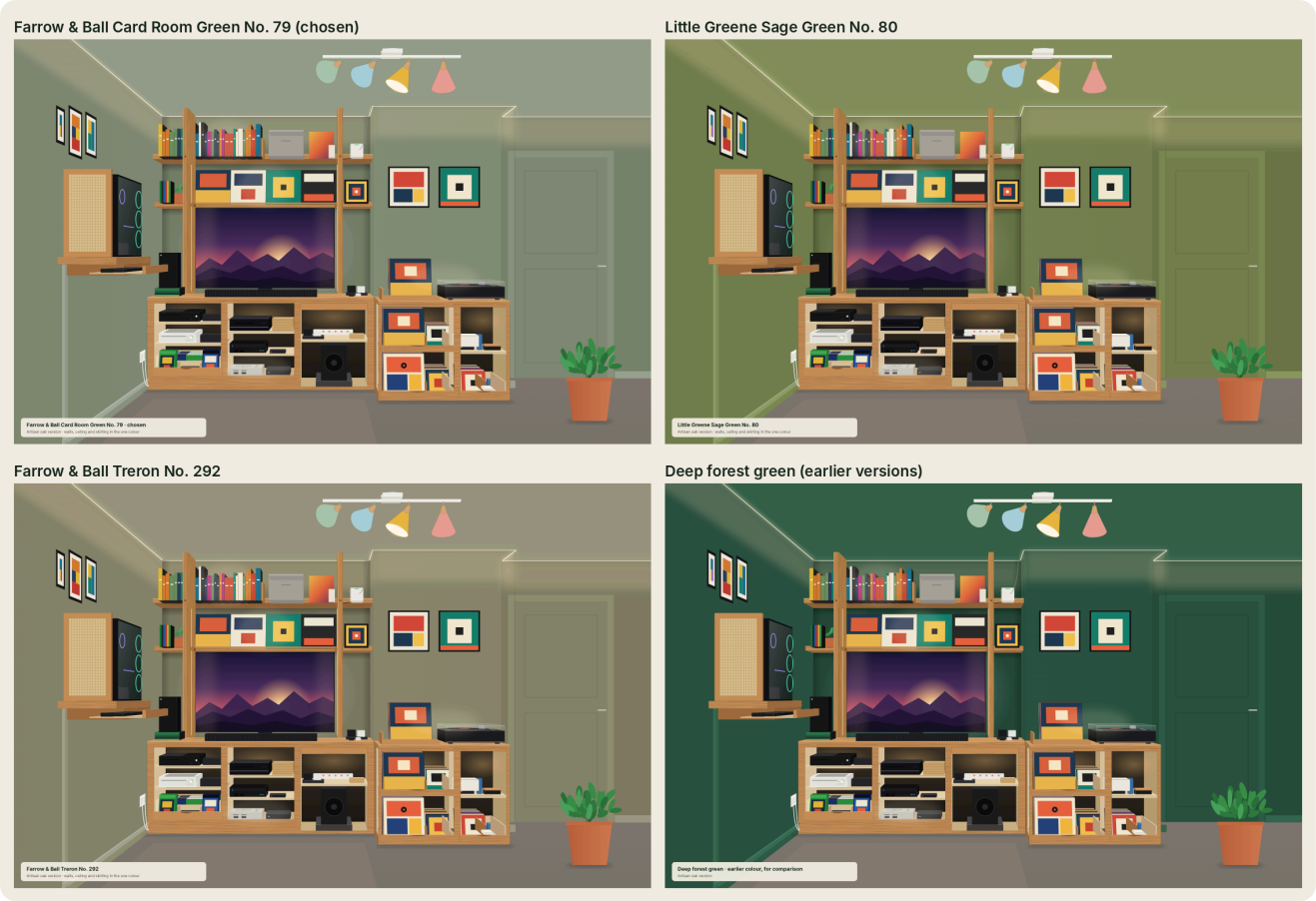
High shelf, centre	Solid oak	128.9 × 30 × 4	Housed 1.2 into both uprights, stub tenons, pegged	1
High shelf, ends	Solid oak	26.45 × 30 × 4	Housed into one upright, on a cleat on the side wall or the reveal	2
Record ledge	Solid oak	128.9 × 7 × 2.4, lip 126.5 × 1.5 × 0.9	Housed and pegged	1
Side shelves	Solid oak	31.75 × 25 × 4 (25.25 + 4.5 through + 2 proud)	Two through-tenons 6 × 2.4, wedged; cleat on the wall	2
Wedges · pegs	Oak · 10 mm oak dowel	6 × 3, tapering 4 to 0.5 mm · 5 long	Glued into saw kerfs · driven from the front edge	4 + spares · 6
Worktop and cabinet top	One solid oak worktop, 300 × 62	186 × 46 and 100 × 44, 3 thick	Chamfered fronts; cut edges sealed	1 board
Face frames	Oak, 20 mm	Base 186 × 73; cabinet 100 × 73; stiles about 5 wide	Glued and pinned; end stiles scribed	2 sets
PC screen	Oak frame 2 × 3 section, pre-woven cane	26 × 48	Cane in a groove with a spline	1

A standard solid oak worktop (3 m long, 62 cm deep) covers both tops: 186 + 100 cm end to end leaves 14 cm, and the long offcut strips make the trays' lips and the browser's rail. Stock worktops come 27 or 40 mm thick: either works; a 40 mm top lifts everything above it by 1 cm.

6 · Paint and light

Paint: Farrow & Ball Card Room Green No. 79

The room is colour-drenched in Card Room Green: walls, ceiling, alcove, chimney breast, skirting, door and frame, radiator and the PC trunking. It's a soft, neutral grey-green that lets the oak and the records lead; the spotlight bar's colours and the sleeves do the contrast work.



The v10 wall in Card Room Green (chosen) beside Sage Green, Treron and the earlier deep green. Screen colours are approximate and the renders are lit as in the evening: judge from painted samples.

- **Finishes:** Estate or Modern Emulsion on walls, ceiling, alcove and breast; Estate Eggshell in the same colour on skirting, door, frame, radiator and trunking (Modern Eggshell if the skirting takes knocks).
- **For the OLED:** a matt mid-tone (LRV about 27) keeps the screen's contrast in the evening. Turn the spots down for films and keep the bias light on.
- **Order:** paint before anything is fixed, so nothing needs cutting in round the uprights, shelves or cabinet.

Skirting: Card Room Green too



- **Recommended: Card Room Green, Estate Eggshell.** Completes the drench, so the room's edges disappear and the oak and records stand out; hides scuffs far better than white.

- **Darker option: Green Smoke** for a defined edge that grounds the room. **Lighter:** a soft off-white rather than brilliant white.
- **Over the white gloss:** wash, key, woodwork primer, then two coats of eggshell; do the door and frame at the same time. Cut the skirting back where the base and the cabinet meet the walls.

Light, in three warm layers

- **The LED strip on the ceiling** (as now): it runs round the room's edge, into the alcove and across the breast. Keep it 2700K on a dimmer. Later, an aluminium channel with an opal diffuser hides the dots, or LED coving throws the light across the ceiling instead.
- **Built into the oak:** a 2700K strip in each upright's groove, on a driver on the fused spur in bay 3; a 6500K bias light on the back of the TV from its USB port.
- **The ceiling light:** a four-head cone spotlight bar in the style of Dunelm's Wolston (£49; 84 × 16 cm, 22.4 cm drop; 4 × E14, 6 W max; IP20), on the existing rose, parallel to the media wall.

Head	Aim
Sage	The books and the left of the high shelf
Pale blue	The record ledge above the TV
Mustard	Down onto the coffee table
Dusty pink	The vinyl cabinet: the deck and the browser **(new)**

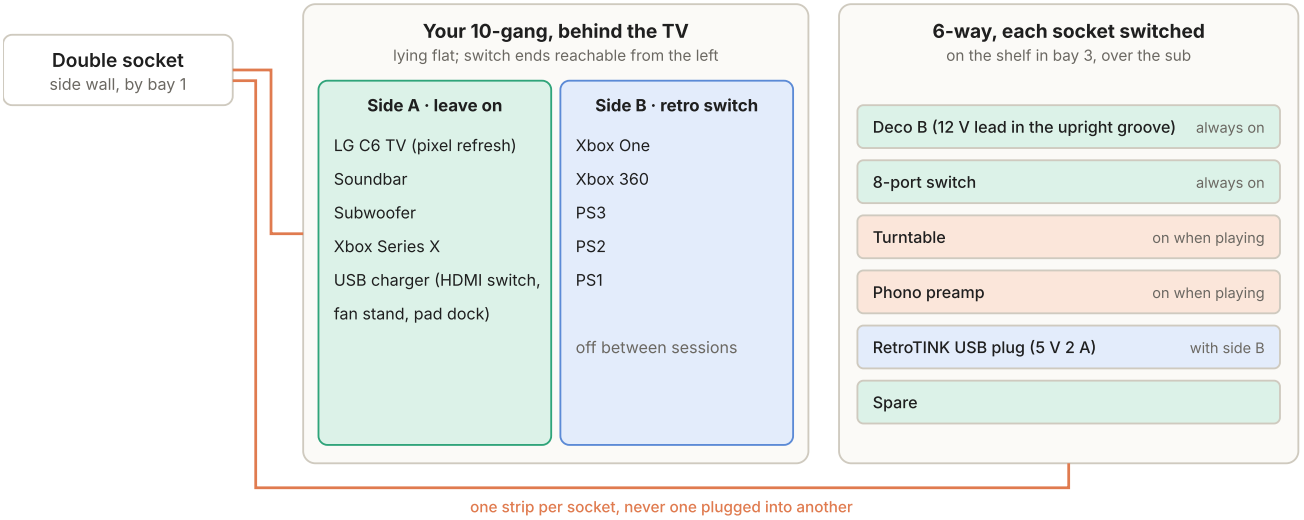
Never point a head at the TV, and tilt the front pair so the shades hide the bulbs from the sofa. Use four warm-white (2700K) dimmable E14 LED bulbs and an LED-compatible dimmer, both fitted on the electrician's visit. With the 235 cm ceiling the lowest shade is at about 212.6 cm.

7 · Power, cables and network

Power

Power map: your 10-gang plus one small switched strip

Fifteen things need power; the 10-gang takes ten, a 6-way with individual switches takes the rest



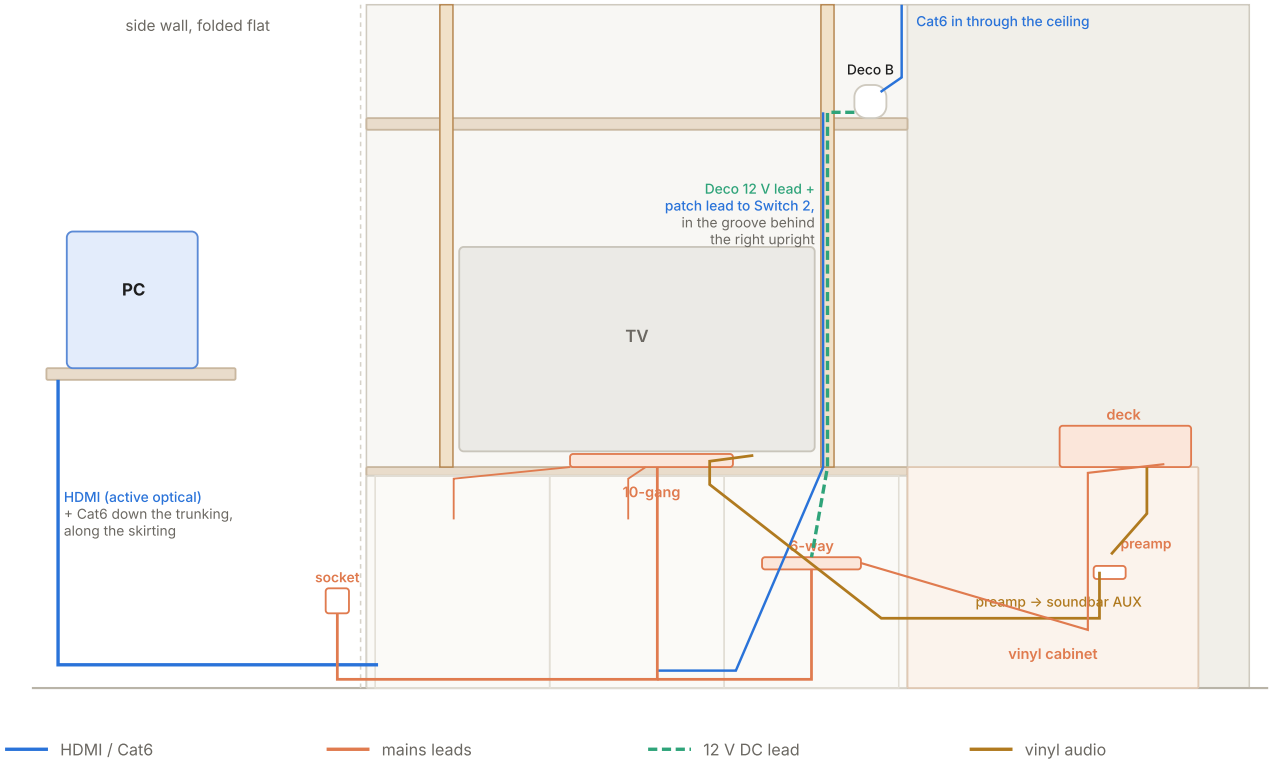
- The socket on the side wall by bay 1 is a single today: swap it for a double, one strip per socket.
- PC: its own wall socket by the PC shelf (it can draw 500–850 W); keep it off these strips.
- Rear speakers: one socket by the sofa.
- The deck and preamp in the vinyl cabinet plug into the 6-way: their leads pass through the back of bay 3 (about 1.5 m).
- Load check: side A about 0.6 kW, side B up to about 0.8 kW with a launch PS3 running; well under the 13 A (3 kW) limit.
- Check the 10-gang is surge-protected; if not, use a surge-protected one for the TV and consoles.
- Label both switch ends: “A: leave on” and “B: retro”.

- **The socket:** on the side wall just in front of bay 1, about 30 cm up, becomes a double: one socket for the 10-gang, one for the 6-way, never one plugged into the other. Moving it 40 cm back into the alcove behind bay 1 hides it completely, if the electrician is there anyway.
- **10-gang:** lying flat on the worktop behind the TV, side B's switch at the left end, reached past the Series X. Side A stays on (TV, soundbar, sub, Series X, USB charger); side B switches the five retro consoles.
- **6-way:** on the shelf in bay 3, at the back. Always on: Deco B and Switch 2. Switched: the turntable, the preamp and the RetroTINK.
- **OLED care:** leave the TV in standby so it runs its pixel refresh.

Cable routes

Cable routes: everything ends in the channel behind the base

Front view, schematic. The side wall is folded flat so the PC's route shows; the vinyl cabinet's leads pass through the back of bay 3

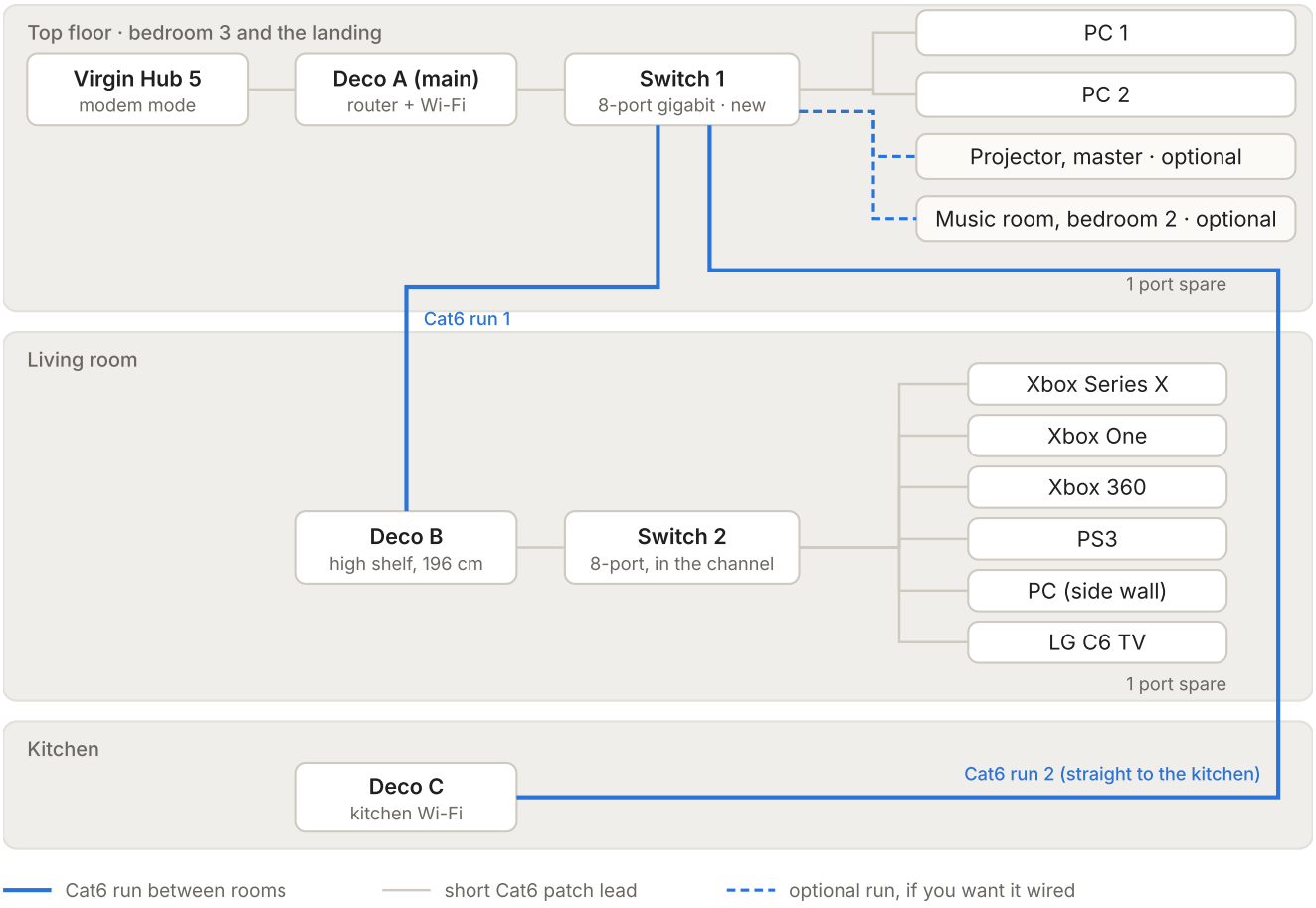


- **The channel:** 4 cm behind the frames, flat two-compartment trunking: mains on one side, HDMI, Cat6 and audio on the other. Switch 2 (15.8 × 10.1 × 2.5 cm) stands on its edge in it.
- **To the TV:** its leads drop through the brush-lined slot in the worktop behind it; no socket behind the TV is needed.
- **Deco B:** the Cat6 comes through the ceiling in the alcove's right corner; its 12 V lead and patch lead run along the back of the high shelf, down the groove behind the right upright and through the worktop to the 6-way and Switch 2.
- **Vinyl cabinet:** its mains and audio leads run along its 2 cm back gap, through the end of the base into bay 3, and on along the channel.
- **Airflow:** console bays stay open-backed; at least 4 cm (ideally 10) of air round the Series X; the power bricks on the free half of their console's row.

Network

Every floor has its own Cat6 link back to bedroom 3

Deco A and Switch 1 live with the Hub 5 and the gaming PCs in bedroom 3; two spare ports can wire the projector and the music room



- **Bedroom 3:** the Hub 5 (in modem mode), Deco A and Switch 1 sit together with the gaming PCs. Switch 1's eight ports: the uplink, two PCs, the two Cat6 runs down to Deco B and Deco C, and, if you want them, the projector in the master and the music room in bedroom 2, leaving one spare.
- **The route down:** from bedroom 3 across the landing to the airing cupboard, which sits over the alcove's corner, and down through its floor, often beside the existing pipes. Run 1 comes through the living room ceiling in the alcove's right corner to Deco B; run 2 follows it down the right upright's groove and through the floor to the kitchen, where Deco C goes on the outside of the brick cupboard, up high.
- **Optional runs:** the master (projector) and bedroom 2 (music room) are both next to the airing cupboard, so short runs along the landing are cheap to add on the same visit. Wired, the projector streams without buffering; the music room gets a solid link for a computer or audio interface.
- **Sloped ceilings:** bedroom 2 and the bathroom slope down to the back wall. Keep any Deco, switch or desk in bedroom 2 on the high side, and run cables along the floor there.
- **Lengths:** roughly 6–9 m for run 1 and 9–13 m for run 2; measure the real route and add 2–3 m to each.

Stage 1, mesh up today: unplug everything from the hub, log in at 192.168.0.1 and turn on Modem Mode (its page then moves to 192.168.100.1); connect Deco A and set it up in the Deco app with the old Wi-Fi name and password; plug Switch 1 into Deco A's other port and the PCs into Switch 1; add Deco B on the high shelf with Switch 2 behind it, and Deco C in the kitchen over Wi-Fi for now; check the Xbox shows NAT: Open.

Stage 2, after the cables: plug each Cat6 run into its Deco's free port, confirm Deco B and C show as "Ethernet" in the app, move each to the centre of its floor, and give the Series X and PC high priority under QoS.

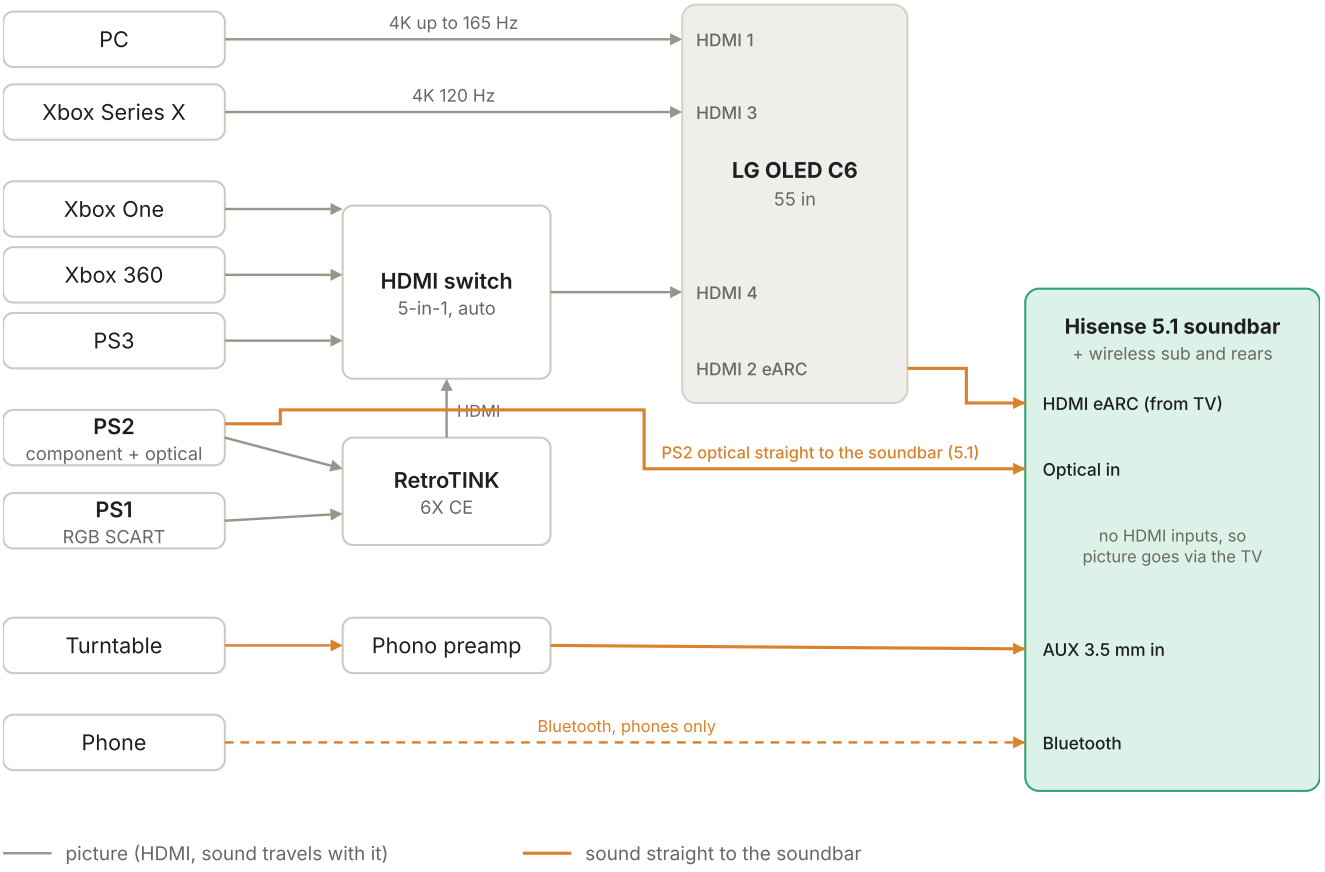
Cabling rules

- Solid pure-copper Cat6, round; not CCA or flat cable for the long runs. Measure the real route and add 2–3 m.
- Cross mains at right angles; never staple through a cable; no kinks; label both ends. Never two cables between the same two devices (a loop takes the network down).
- **Done when:** Deco B and C show "Ethernet", each PC shows 1.0 Gbps, the Xbox shows NAT: Open, and phones hold a strong signal in the kitchen and across the living room.

AV: everything sounds through the Hisense

Everything sounds through the Hisense: directly where it can, via the TV where it can't

The soundbar has no HDMI inputs, so HDMI sources reach it over eARC; vinyl, PS2 audio and phones go straight in



Source	Picture	Sound
PC · Xbox Series X	TV HDMI 1 · HDMI 3	Via the TV over eARC (5.1 or Atmos)
Xbox One, Xbox 360, PS3	HDMI switch → TV HDMI 4	Via the TV over eARC, 5.1
PS2	Component → RetroTINK → switch → TV	Optical straight to the soundbar, 5.1 where the game has it
PS1	RGB SCART → RetroTINK → switch → TV	Stereo through the RetroTINK and eARC
Turntable · phone	—	Preamp → AUX · Bluetooth

- **TV:** rename the inputs (HDMI 4 "Retro"), Deep Colour on HDMI 1, 3 and 4, Game Optimiser on the console inputs, sound out to HDMI eARC with digital audio on Auto or Passthrough. PS3, 360 and Xbox

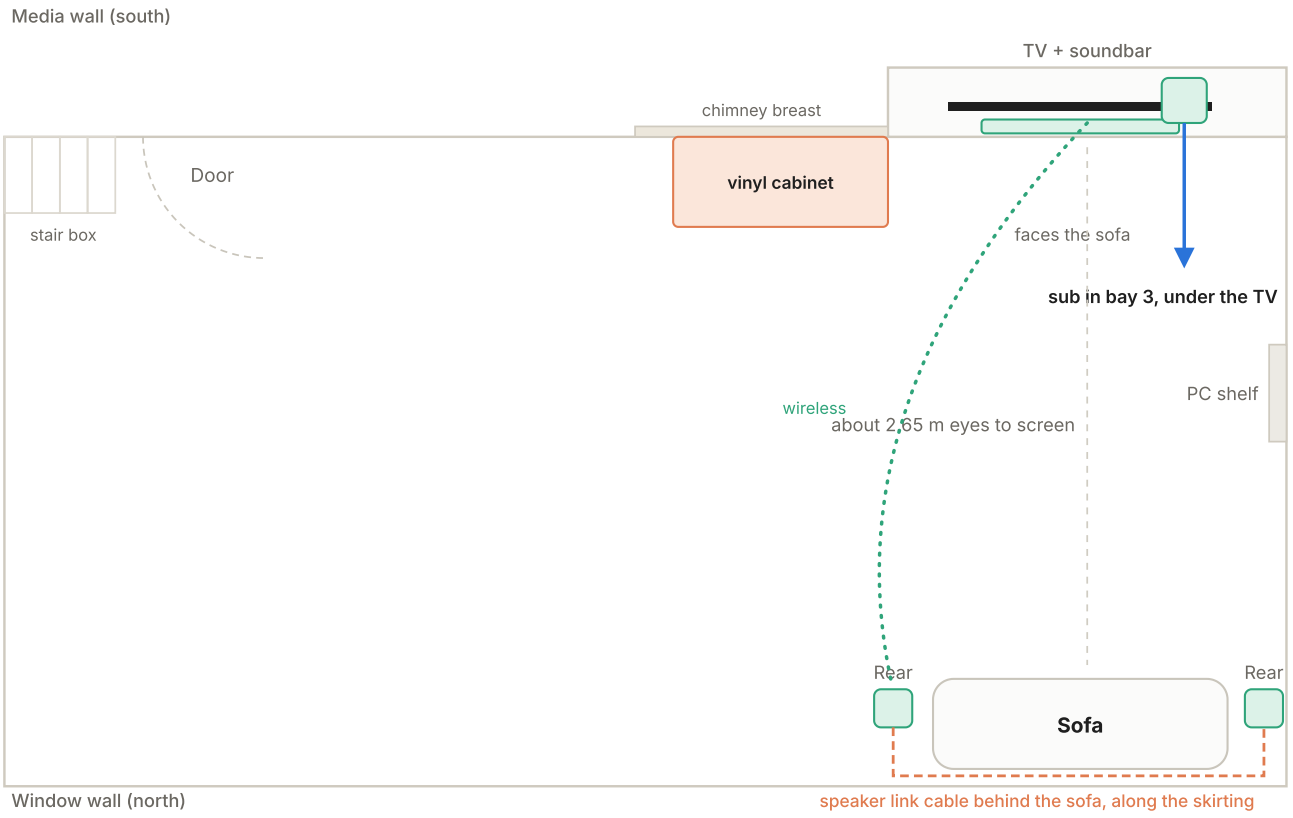
One output 5.1 over HDMI; PS3 "Control for HDMI" off.

- **RetroTINK-6X CE:** motion-adaptive deinterlacing suits the PS2's mostly interlaced games, up to 1440p for the TV to finish to 4K. It needs a 5 V 2 A USB plug (none in the box) on its own switched socket of the 6-way. Set the PS2's Component Video Out to Y Cb/Pb Cr/Pr, choose 60 Hz where a game offers it, and use RGB SCART for the PS1.
- **Sharper still:** PCSX2 on the PC renders PS2 games at native 4K from your own discs and BIOS.

Sound layout

The sub fires down the room at the sofa

From above, not to scale, media wall at the top. The sub sits in bay 3 under the TV; the rears share one socket and a cable behind the sofa



- **Sub:** in bay 3 under the TV, on its riser, facing the sofa; turn it down rather than moving it if the bass booms.
- **Rears:** either side of the sofa at about seated ear height, angled in; on the AX5100G they're linked by a cable, so one socket and a run behind the sofa. The left one can hang on the side wall by its keyhole.
- **Tuning:** set the sub and rear levels from the sofa with the soundbar's remote.

8 · Build guide

Sizes are finished sizes in centimetres. Walls are rarely square: measure the alcove's width at the floor, 76, 156, 196 and the ceiling, and cut the worktop and shelves to the smallest.

What it's made of

Element	Made from	Finished size	Fixing
Plinth (bays 1–2)	18 mm ply ladder frame	120 × 38 × 9	Shimmed level; frames screwed down onto it
Bays 1–2	2 × IKEA BESTÅ frame, back panels off	60 × 40 × 64 each, tops at 73	Joined, 4 cm spacers to the wall, 2 anti-tip brackets each
Bay 3	18 mm birch ply, back open	60 × 40 × 73	Bolted to bay 2, 4 cm spacers, 2 anti-tip brackets
Internal shelves	4 × BESTÅ shelf (20 kg each); 1 ply shelf in bay 3	56 × 36	On pins at the fit-proof heights
Alcove worktop	Solid oak	186 × 46 × 3, top 76	Screwed up through the tops in slotted holes; brush-lined slot behind the TV
Uprights	Solid oak	159 × 30 × 4.5	Two 10 mm dowels into the worktop, two concealed brackets each
Side shelves	Solid oak	31.75 × 25 × 4, top 156	Wedged through-tenons, cleat on the wall
Record ledge	Solid oak	128.9 × 7 × 2.4, top 155.4	Housed and pegged
High shelf	Solid oak, three lengths	30 × 4, top 196	Housed and pegged; end lengths also on cleats
Vinyl cabinet	18 mm birch ply carcass, oak face frame, end panel and top	100 × 44 × 76	Anti-tip brackets into the breast
Record trays · singles drawer	9 and 12 mm birch ply, oak dividers and lips	53.9 × 40 · 33.9 × 40	400 mm runners into the ply
Game drawer · hidden drawer	Birch ply, oak lips	54.3 × 35 · 52 × 30 × 6	350 mm · 300 mm runners
PC shelf box	27 mm oak top, 18 mm oak sides	65 × 32, box 9 deep	French cleat and 2 steel brackets
Face frames	20 mm oak	186 × 73 and 100 × 73	Glued and pinned; end stiles scribed

Tools

- Drill-driver, SDS or hammer drill with masonry bits; spirit level (a cross-line laser makes the heights quick); tape, combination square, cable detector.
- Track or circular saw for the worktop (or have the merchant cut the ply and oak to the cut lists), jigsaw for notches.
- For the joints: sharp chisels (6, 12, 25 mm), marking gauge, tenon saw, router with a straight bit for the housings and grooves, 10 mm bit for the pegs, mallet; a cove bit for the finger pulls.
- Clamps, sander, wood glue; dust mask, eye and ear protection.

Order of work

1. **Measure and check:** the alcove's width at five heights, plumb and level, the ceiling, the door's swing, and your records and games.
2. **Network stage 1** (any time): hub in modem mode, Decos and switches set up over Wi-Fi.
3. **Electrician:** the Cat6 runs from bedroom 3 (and the optional ones), the double socket, sockets by the sofa and under the PC shelf, the fused spur, the spotlight bar and dimmer. Mark every cable route before painting.
4. **Paint** the whole room in Card Room Green, skirting and door in eggshell, and let it cure for a few days.
5. **Mark the heights** with a level or laser: 73 (worktop), 152 (side shelf undersides), 153 (ledge), 192 (high shelf underside), and the uprights' lines at 25.25–29.75 and 156.25–160.75 from the side wall.
6. **Base:** the plinth, then the frames on it, then bay 3; spacers, anti-tip brackets, the trunking in the channel, and the cables run in.
7. **Worktop:** scribe, oil all over (underside too), screw up through the tops in slotted holes, cut the slot behind the TV.
8. **Inside the base:** shelves at the fit-proof heights, the game drawer, bay 3's shelf, the hidden drawer, the sub on its riser.
9. **Oak joinery:** dry-fit each upright with its shelves on the bench; lay the Deco leads in the right upright's groove; stand the uprights on their dowels, fix the brackets, fit the shelves, ledge, wedges and pegs.
10. **Vinyl cabinet:** build it, check the door, fix it to the breast, fit the trays and the singles drawer, then the face frame, end panel and top.
11. **PC shelf:** the box, the tray, the cleat and brackets, the cane screen, the trunking to the skirting.
12. **Lighting:** LED in the upright grooves, the ceiling strip through its notches, the bias light, then aim the spot bar.
13. **Connect:** place everything, network stage 2, label every plug and HDMI input, and test.

9 · Buy list and tracker

Every purchase and job, in build order, comes to about £2,207–4,249 plus the electrician's quote. Estimates are approximate UK prices; changed and new items are marked. Tick items off and write in what you paid.

Phase	Estimate (£)	Biggest item
1 Network, now	51–70	two 8-port switches
2 Electrician and cabling	130–240, plus the electrician	Cat6 and the switched 6-way
3 Paint	220–540	paint
4 Base unit	322–682	BESTÅ frames and the PC shelf
5 Oak joinery	705–1,320	the solid oak worktop
6 Vinyl cabinet	325–570	record trays and the carcass
7 Lighting	100–250	spotlight bar and LED
8 TV, consoles and AV	283–410	RetroTINK-6X CE
9 Vinyl kit	71–167	phono preamp
10 Finishing	—	—
Total	**2,207–4,249, plus the electrician**	

Already owned: Virgin Hub 5, three Deco X1500s, two TL-SG105S switches (both become spares), LG OLED C6 55" and its stand, a 10-gang with a switch per side, Hisense AX5100G 5.1 soundbar, the Series X fan stand, the turntable and all the consoles.

Status	Item	Qty	Estimate (£)	Paid (£)
1 NETWORK, NOW				
☐ To do	Put the Virgin Hub 5 into modem mode			
☐ To do	Set up Deco A, Switch 1 and the PCs in bedroom 3			
☐ To do	Add Deco B and Deco C over Wi-Fi for now; speed-test each room and the Xbox			
☐ To buy	Cat6 patch leads, 0.5–2 m	11	15–20	
☐ To buy	TP-Link TL-SG108 8-port switch: one for the living room, one to replace the 5-port in bedroom 3	2	36–50	
2 ELECTRICIAN AND CABLING				
☐ To do	Book one electrician visit: Cat6 from bedroom 3 across the landing and down through the airing cupboard, one run to the alcove's right corner (Deco B) and one on to the kitchen (Deco C), a spare pulled with each; if you want them, runs to the master (projector) and bedroom 2 (music room); swap the single socket on the side wall in front of bay 1 for a double (or move it into the alcove behind bay 1); a socket by the sofa for the rear speakers if there isn't one; a socket under the PC shelf; a fused spur for the LED drivers; fit the spotlight bar and an LED-compatible dimmer **(changed: route from bedroom 3, optional runs)**		Quote	
☐ To buy	Cat6 solid pure copper, round (not CCA): two runs, four with the optional ones	2–4 runs	30–60	
☐ To buy	6-way surge-protected extension with a switch per socket: Deco B, Switch 2, turntable, preamp, RetroTINK	1	25–40	
☐ To buy	4-port USB charger, 5 V 2 A per port (HDMI switch, Xbox fan stand, pad dock)	1	10–20	
☐ To buy	USB plug, 5 V 2 A or more, for the RetroTINK on the switched strip	1	5–10	
☐ To buy	Cable kit: brush grommets, velcro ties, labels	1	20–30	
☐ To buy	Flat two-compartment trunking for the 4 cm channel behind the base, about 2 m **(changed: flat)**	1	10–20	
☐ To buy	Slim paintable trunking for the PC drop and the skirting run, about 2.5 m	1	15–30	
☐ To buy	DC extension lead for the Deco adapter, only if its lead won't reach the 6-way in bay 3 (match the plug)	1	5–10	
☐ To buy	PC tidy kit: braided sleeve, right-angle C13 lead, shelf-notch grommet	1	10–20	
3 PAINT				
☐ To do	Sample Farrow & Ball Card Room Green No. 79 on an A4 board; check it by day and with the LED strip on			
☐ To buy	Card Room Green: Estate (or Modern) Emulsion for walls, ceiling, alcove and chimney breast; Estate Eggshell for skirting, door, frame and trunking; primer and undercoat		200–500	
☐ To buy	Woodwork primer, sanding pads and sugar soap for the white skirting and door	1	20–40	
☐ To do	Paint the whole room before anything is fixed: alcove and breast included; skirting and door from white to eggshell			
4 BASE UNIT				
☑ Done	Check the alcove back wall for fixings (solid)			
☐ To do	Measure the alcove's width at the floor, 76, 156, 196 and the ceiling, and the floor's level across it **(new)**			
☐ To do	Check the side wall where the PC shelf goes: solid, or plasterboard on studs			

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To buy	Two IKEA BESTÅ 60×40×64 frames, white stained oak effect, for bays 1–2	2	70–200	
☐ To buy	Plinth for bays 1–2: 18 mm ply ladder frame 120 × 38 × 9 cm, plus shims	1	15–30	
☐ To buy	Bay 3 carcass for the sub: 18 mm birch ply, 60 × 40 × 73 cm from the floor, a shelf, connector bolts **(changed: holds the sub now)**	1	50–90	
☐ To buy	BESTÅ shelves, 56 × 36 cm (20 kg max each), two each for bays 1 and 2	4	12–32	
☐ To buy	Anti-tip brackets and 4 cm spacer blocks: both frames, bay 3 and the vinyl cabinet **(changed)**	1 set	20–30	
☐ To do	Measure about ten games of each kind (PS2, Xbox 360, Xbox One) before cutting the game drawer			
☐ To buy	Game drawer kit: 12 mm birch ply cut to size, 350 mm full-extension soft-close runners (35–49 kg), oak lip, follower block, felt	1	50–90	
☐ To buy	Hidden slim drawer for bay 3: oak front 56.4 × 7.4 cm, 12 mm ply box 52 × 30 × 6 cm, 300 mm runners **(changed: wider)**	1	30–50	
☐ To buy	Compact isolation riser for the sub, about 30 × 35 cm, 4–6 cm tall	1	20–40	
☐ To buy	PC shelf box: 27 mm oak top 65 × 32, 18 mm oak sides, keyboard tray on 300 mm runners, French cleat, two steel brackets rated 25 kg+	1	55–120	
☐ To do	Build the plinth and bay 3, fix everything to the wall, fit the shelves and drawers; place the TV, soundbar, Series X and sub			
5 OAK JOINERY				
☐ To buy	Two solid oak uprights, 165 × 30 × 4.5 cm (cut to 159), quarter-sawn if possible **(changed: longer)**	2	150–300	
☐ To buy	Solid oak for the shelves: high shelf in three lengths (26.45, 128.9, 26.45 × 30 × 4), two side shelves 31.75 × 25 × 4, record ledge 128.9 × 7 × 2.4 **(changed: no turntable shelf)**	1 set	170–300	
☐ To buy	Oak wedges, 10 mm oak dowel for pegs and dowels, concealed upright brackets	1	15–30	
☐ To buy	One solid oak worktop, 300 × 62 cm, full stave: cut into the alcove top (186 × 46) and the cabinet top (100 × 44), chamfered fronts; offcuts for lips and the browser's rail **(changed)**	1	250–450	
☐ To buy	Oak face frames, 20 mm: 186 cm for the base, 100 cm for the vinyl cabinet **(changed)**	1	100–200	
☐ To buy	Cane screen for the PC: oak frame and pre-woven cane, 26 × 48 cm	1	20–40	
☐ To do	Cut the joints and fit the uprights, shelves, ledge and face frame; notch the uprights for the ceiling strip			
6 VINYL CABINET				
☐ To buy	Vinyl cabinet carcass: 18 mm birch ply for bay A (60 × 40 × 73) and bay B (40 × 40 × 73) with a shared middle side, bay B's shelf, connector bolts; oak-veneered end panel for the side you see **(new)**	1	90–160	
☐ To buy	Two face-out record tray kits: 12 and 9 mm birch ply cut to size, 6 mm oak dividers, 15 mm oak lips, raked rests, felt, two pairs of 400 mm runners rated 68–70 kg	2	150–240	
☐ To buy	Singles drawer for bay B: ply box 33.9 × 40 × 13, oak divider and lip, 400 mm runners rated 45 kg+ **(new)**	1	40–70	
☐ To buy	LP browser for the cabinet top: 18 mm oak base 44 × 24, low front rail and one side	1	15–35	
☐ To buy	Isolation feet for the turntable	1	10–25	
☐ To buy	Two LP frames for the chimney breast, hung from 150 cm **(new)**	2	20–40	

Status	Item	Qty	Estimate (£)	Paid (£)
☐ To do	Build the cabinet, check the door clears it, fix it to the breast with anti-tip brackets, fit the trays and the singles drawer			
7 LIGHTING				
☐ To buy	LED for the upright grooves: 2700K strip, two aluminium channels, driver; 6500K bias light for the TV	1	40–90	
☐ To buy	Ceiling light: Dunelm Elements Wolston 4-light spotlight bar (£49), plus four warm-white (2700K) dimmable E14 LED bulbs, 6 W max	1	60–80	
☐ To buy	Later: aluminium channel and opal diffuser for the existing ceiling LED strip (optional)	1	0–80	
8 TV, CONSOLES AND AV				
☐ To buy	5-in-1 4K60 HDMI switch with remote	1	20–35	
☐ To do	Measure the PC-to-TV cable route (about 3–4 m)			
☐ To buy	Active optical HDMI cable for the PC, certified Ultra High Speed (48 Gbps), about 5 m	1	20–60	
☐ To buy	High Speed HDMI leads, 1–2 m	5	20–30	
☐ To buy	RetroTINK-6X CE upscaler for the PS2 and PS1 (order on restock; UK price includes VAT and import fees)	1	200–240	
☐ To buy	PS2 component video cable, good quality	1	10–20	
☐ To buy	Optical (TOSLINK) cable, 2 m, PS2 straight to the soundbar	1	5–10	
☐ To buy	RGB SCART cable for the PS1	1	8–15	
☐ To do	Connect the TV inputs; set up the PS2 and RetroTINK			
9 VINYL KIT				
☐ To buy	Replacement stylus for the turntable (exact match for its model)	1	15–40	
☐ To buy	External phono preamp (only if the deck has a PHONO/LINE switch)	1	40–100	
☐ To buy	RCA-to-3.5 mm lead, 3 m: preamp in the cabinet to the soundbar's AUX **(changed: longer)**	1	6–12	
☐ To buy	Carbon-fibre record brush	1	10–15	
☐ To do	Set the tracking weight; soundbar in Music mode			
10 FINISHING				
☐ To do	Turn on HDMI-CEC (SIMPLINK); PS3 Control for HDMI off			
☐ To do	Deco app: prioritise the Series X and PC, guest Wi-Fi, automatic firmware			
☐ To do	Label each switched socket and HDMI switch input with the console's name			
☐ To do	Set the sub and rear speaker levels from the sofa			

10 · Still to confirm, later ideas, sources

Still to confirm

- ☐ The vinyl corner moving to an oak cabinet on the chimney breast (because the alcove is 39 cm narrower than assumed).
- ☐ The door to the hall: which side it hinges, and that it opens clear of the cabinet's west end (17.5 cm from the breast's corner).
- ☐ The west end's sizes: which of 935, 1045, 660 and 820 are the recess, the door and the stair box.
- ☐ The turntable's model (the top allows up to 45.2 × 35.2 cm) and whether it has a PHONO/LINE switch.
- ☐ The Series X fan stand's model: the left gap fits a footprint up to about 24 cm.
- ☐ The PC case's model, and that its window is on the left (so the back faces the sofa).
- ☐ The 10-gang's length and whether it's surge-protected.
- ☐ The 23 × 22 cm box in the window's west corner.
- ☐ Wired ports for the projector (master) and the music room (bedroom 2)?
- ☐ The RetroTINK's size when it arrives (bay 2's bottom row reserves about 30 cm beside the PS1).
- ☐ Records: roughly how many LPs and singles, and are most in thick outer sleeves?
- ☐ Games: roughly how many PS2, Xbox 360 and Xbox One? Past about 48, plan overflow.
- ☐ The floor in the alcove and at the breast: carpet or hard floor (the plinth and cabinets must sit level).
- ☐ The side wall where the PC shelf goes: solid or plasterboard?
- ☑ ~~The real room: alcove 186 × 38, chimney breast 117.5, ceiling 235, skirting 60 × 20.~~
- ☑ ~~Artisan oak is the main build; Card Room Green with matching eggshell skirting; the Wolston-style spot bar on the existing rose.~~
- ☑ ~~Every record face-out; the game drawer as designed; the LED strip on the ceiling stays.~~
- ☑ ~~Power: the side-wall socket in front of bay 1 becomes a double.~~

Later ideas

- **Dave and Emma's watch log:** a shared film and TV log, like a Spotify Blend for what you watch: log each film or episode, mark whether you watched it together, rate it, and later get suggestions for what to watch next. It could live on the TV's browser or your phones. To be shaped up separately.

Sources

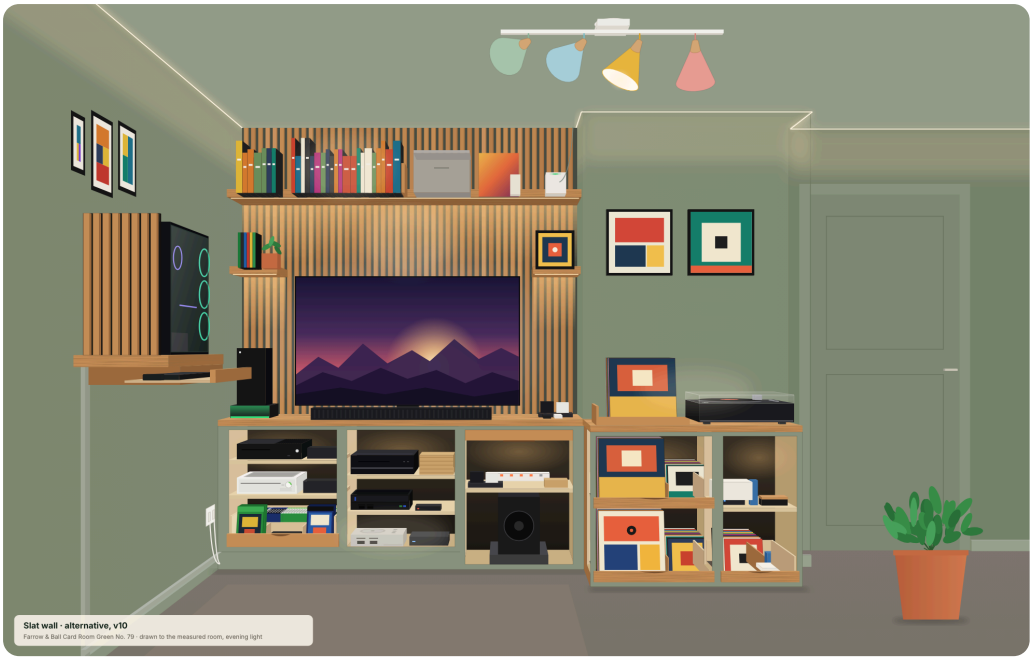
Numbered sources are the item sizes used in the fit proof (section 3).

1. Your hand-measured sketch of the living room, in mm (alcove 1860 × 380, chimney breast 1175, ceiling 2350)
2. LG: OLED55C6 specifications: 1222 × 703 × 45.1 mm without stand, 1222 × 757 × 230 mm with stand, stand 470 × 230 mm, 14.1/15.9 kg, VESA 300 × 200 (lg.com, OLED55C6 product pages)
3. Trusted Reviews and Expert Reviews, Hisense AX5100G reviews: soundbar 920 × 60 × 90 mm, subwoofer 214 × 294 × 304 mm (W × H × D) with a rear-firing port, rear speakers 90 × 144 × 109 mm
4. Wikipedia, Xbox Series X: 151 × 151 × 301 mm; Pure Xbox (Nov 2020) quoting Microsoft: at least 4 cm clear above the console on an open stand, 10–15 cm recommended on all sides
5. IKEA: BESTÅ frame 60 × 40 × 64 cm (10247379); BESTÅ shelf 56 × 36 cm, 20 kg maximum load
6. TP-Link: TL-SG108 specifications, 158 × 101 × 25 mm (tp-link.com)
7. Laptop Outlet: TP-Link Deco X1500 3-pack listing, 110 × 110 × 114 mm per unit
8. Audio-Technica: AT-LP120XUSB specifications, 452 × 352 × 141.6 mm, 8 kg without dust cover (audio-technica.com)
9. RetroTINK: RetroTINK-6X CE shop page (power 5 V 2 A by USB; no size published)
10. UGREEN: 5-in-1 4K60 HDMI switch CM568 (90512), 127 × 59 × 25 mm
11. Original Xbox One power supply: 17 × 7.5 × 4.8 cm (6.69 × 2.96 × 1.88 in), as listed by several replacement-PSU sellers

12. Wikipedia: Comparison of PlayStation consoles (PS1 270 × 188 × 60, PS2 302 × 183 × 78, PS3 325 × 274 × 98 mm); Xbox One (333 × 274 × 79 mm); Neowin: official Xbox 360 dimensions (309 × 258 × 83 mm)
13. Record and case sizes: LP sleeve 31.4 cm (12⅜ in); 7-inch sleeve 18.3–18.4 cm; DVD-style keep case 19 × 13.5 × 1.4 cm; Xbox One case 17 × 13.5 × 1.1–1.2 cm (videogamecasesize.info; Walvis Products; Wikipedia: Keep case)
14. Dunelm: Elements Wolston 4-light spotlight bar, £49: 84 × 16 cm, 22.4 cm drop, shades 15 cm, E14 6 W max (dunelm.com, product 1000189449)
15. Emuca heavy-duty full-extension runners, 400 mm, 68 kg per pair; Accuride DZ3832EC 350 mm, 43 kg; runner fitting allowance 12.5 mm a side
16. Xbox 360 original 203 W power supply: no manufacturer size published; allowance of 21 × 10 × 6 cm used here (measure yours)
17. Virgin Media: Hub 5 modem mode
18. RetroTINK: Introducing the RetroTINK-6X CE (inputs including TOSLINK, deinterlacing); RetroTINK-2X Pro listed as legacy; RetroTINK-4K CE
19. Expert Reviews: LG C6 OLED review (4 × HDMI 2.1, one eARC, ports on the left); JB Hi-Fi: LG C6 specifications (165 Hz mode for PC sources)
20. Hisense AX5100G spec sheet (inputs, auto power-on); JB Hi-Fi Business: AX5100G (rears linked by cable, wireless to the soundbar)
21. TP-Link: Deco X1500 specifications (two gigabit ports per unit)
22. Glorious Vinyl Vault specifications (80 LPs and 24.5 kg per drawer), a check on the tray capacity
23. Screwfix: Essentials 964VN 350 mm runners; Metrol: Accuride DZ3832EC; IKEA: BESTÅ drawer runners
24. NIST SP 500-252 / CLIR pub121: Care and Handling of CDs and DVDs (store upright, book style)
25. Can-Am, mmdesign (DiscSox): spine-up media drawer capacities
26. Sony PS3 and Microsoft Xbox 360 S manuals: ventilation; IKEA UK: secure it
27. Ideal Home: colour drenching; The Paint Shed: colour drenching with Farrow & Ball; Livingetc: sage green shades (Card Room Green, Treron, Sage Green, Green Smoke)
28. Pontata: Card Room Green vs Little Greene Sage Green (approximate screen colours)
29. Homebase and NC Tiles: Farrow & Ball Card Room Green No. 79 in Modern Eggshell and Estate Eggshell
30. haart Medway: floor plans and listing photos for the house (Portsmouth Close)
31. ConsoleMods: PS2 Component Video Out setting

Appendix · The slat wall alternative

Kept for reference, now drawn to the measured room. The base, the vinyl cabinet, the records and the consoles are identical; above the worktop, oak slat panels on battens run from the worktop to the ceiling instead of the uprights, the shelves sit on hidden cleats, the face frame is painted Card Room Green, and the light comes from strips under each shelf. Quicker to fit and good for sound, but it follows a current trend, and it loses the record ledge and the joints on show.



Options we rejected along the way

Option	Why not
Hub as router, Decos as access points	Loses HomeShield and parental controls
All-wireless mesh · powerline	Speed drops at each hop · slower and less stable than Cat6
2.5 Gb switches	The Decos and PCs top out at 1 Gb
Games two deep, or lying flat	The back row is hidden; flat stacks are hard to pick from
Records spine-up in one drawer	Showed only a handful of covers; face-out trays show every one
A green feature wall only · white skirting	The drench ties the room together; white cuts a bright line round it
A paper lantern or long pendant	Replaced by the spot bar you chose
RetroTINK-2X Pro · cheap PS2-to-HDMI adapters	Legacy, no 480p · soft, washed-out picture on a 55" OLED
Bluetooth from the turntable	Compresses the sound; the AUX lead keeps full quality